

KTH ROYAL INSTITUTE
OF TECHNOLOGY

Doctoral Thesis in Machine Design with specialization in Integrated Product Development

Circling the Squares

Radical Innovation and Management Control Systems in the Circular Economy

JOHAN AREKRANS

Stockholm, Sweden 2023

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Abstract

An economic system built upon the sale of tangible goods produced using finite resources is inherently unsustainable. The idea of a Circular Economy (CE) as a viable pathway to sustainability has gained considerable attention from academia, policymakers, and the business community. The CE builds on a new logic in the business environment in which industrial firms operate that requires substantial changes. We face a significant knowledge gap concerning how firms internally manage the implementation and adoption of CE. Filling in this knowledge gap is critical to understanding how to implement CE at large and established firms with a considerable history of a linear logic, which are known to favor incremental improvements to existing business opportunities over radical innovation.

This thesis aims to delve into the managerial challenges that large, established industrial firms encounter when implementing CE principles, and how they address these challenges, particularly through managerial controls. Additionally, it aims to provide insights into how management can facilitate radical circular innovation and support the circular transformation of incumbent firms. To accomplish this aim, existing knowledge on barriers to CE has been synthesized through a systematic literature mapping and a systematic literature review. Next, interviews (n=68) at five large and established industrial firms provide empirical insights concerning the implementation and adoption of CE principles. In addition to the four appended papers, this comprehensive summary theorizes about the empirical findings using literature on radical innovation, sensemaking, and managerial controls.

Building on the empirical papers, this analysis sheds light on two distinct patterns in how managers frame CE: incremental framing versus radical framing. It is argued that this has considerable consequences concerning with respect to ambitions, operationalization, and the means used to achieve a CE. In addition, the thesis theorizes about the relationship between the management control systems and the prevailing framings of CE. Based on an understanding of this relationship and the tensions and conflicts associated with the different framings, it is argued that managerial controls can act both as barriers to and enablers of CE adoption at incumbent firms. In addition to the practical and theoretical implications of this finding, the thesis pinpoints limitations and assumptions concerning managerial control systems in relation to CE and propose new avenues in light of this.

Finally, the analysis is synthesized into an integrative framework that differentiates three different modes of organizational behaviors within the context of CE transitions: optimizing, transforming, and systems building. This framework integrates the theoretical foundations of the thesis, acknowledges the identified issues, and offers actionable implications for researchers and practitioners.

Sammanfattning

Ett ekonomiskt system byggt på försäljning av varor tillverkade av ändliga resurser är i grunden ohållbart. Idén om en Cirkulär Ekonomi (CE) har vuxit fram som en möjlig väg framåt och har fått betydande uppmärksamhet från akademin, beslutsfattare och näringslivet. CE bygger på en ny logik i affärsmiljön där industriella företag verkar och kräver betydande förändringar. Det finns dock en betydande kunskapslucka om hur företag hanterar implementering och antagande av CE internt. Denna kunskapslucka är särskilt avgörande att förstå i stora och etablerade företag med en lång historik av en linjär logik, och som är kända för att gynna inkrementella förbättringar av befintliga affärsmöjligheter framför radikalt nya.

Denna avhandling syftar till att fördjupa sig i de ledningsmässiga utmaningarna som stora och etablerade industriföretag står inför när de implementerar CE-principer och hur de hanterar dem, särskilt genom kontrollmekanismer. Dessutom syftar den till att ge insikter om hur ledningen kan underlätta radikal cirkulär innovation och stödja en cirkulär omvandling av befintliga företag. För att uppnå detta mål har befintlig kunskap om hinder för CE syntetiserats genom en systematisk litteraturkartläggning och en systematisk litteraturstudie. Därefter ger intervjuer (n=68) i fem stora och etablerade industriella företag empiriska insikter om implementering och antagande av CE-principer. Förutom de fyra bifogade artiklarna teoretiserar denna kappa kring de empiriska resultaten med hjälp av litteratur om radikal innovation, meningsskapande och kontrollmekanismer.

Med utgångspunkt i de empiriska artiklarna så visar analysen två olika tolkningar eller ”inramningar” av CE inom företagen: en inkrementell inramning och en radikal inramning. Det argumenteras för att inramningen har betydande konsekvenser när det gäller ambitioner, operationalisering och tillvägagångssätt för att nå CE inom företaget. Dessutom teoretiserar avhandlingen om förhållandet mellan kontrollmekanismer och vilken inramning av CE som blir dominerande inom företaget. Genom att förstå detta förhållande och de spänningar och konflikter som följer olika ramverk, hävdas det att ledningskontroller kan fungera både som hinder och effektiva mekanismer för CE-antagande i befintliga företag. Förutom de praktiska och teoretiska innebörderna pekar avhandlingen på begränsningar och antaganden kring kontrollmekanismer i relation till en cirkulär logik, och föreslår nya forskningsspår i ljuset av detta.

Slutligen syntetiseras analysen till ett integrerat ramverk med tre olika mönster av organisatoriskt beteende i en övergång till CE: *Optimering*, *Transformation* och *Systembyggande*. Detta ramverk integrerar de teoretiska grunderna för avhandlingen, pekar på de identifierade problemen, och beskriver innebörden för forskning och näringsliv.

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Thank you!
Stockholm, November 2023
Johan Arekrans

List of Appended Papers

Paper A

Sopjani, L., **Arekrans**, J., Laurenti, R., & Ritzén, S. (2020). Unlocking the linear lock-in: Mapping research on barriers to transition. *Sustainability*, *12*(3), 1034. https://doi.org/10.3390/su12031034

Paper B

Arekrans, J., Sopjani, L., Laurenti, R., & Ritzén, S. (2022). Barriers to access-based consumption in the circular transition: A systematic review. *Resources, Conservation and Recycling*, *184*, 106364. https://doi.org/10.1016/j.resconrec.2022.106364

Paper C

Arekrans, J., Ritzén, S., & Laurenti, R. (2022). The role of radical innovation in circular strategy deployment. *Business Strategy and the Environment*, *32*(3), 1085–1105. https://doi.org/10.1002/bse.3108

Paper D

Arekrans, J., Ritzén, S., & Nilsson, S. (2023) Tensions in Management Controls; Enabling radical innovation for a Circular Economy. [Submitted manuscript]

List of Additional Publications during Doctoral Studies

Sopjani, L., Laurenti, R., **Arekrans**, J. & Ritzén, S. (2019). A systematic map for classifying studies on barriers related to circular economy, sharing economy, collaborative consumption, and product-service systems. Paper presented at 19th European Roundtable for Sustainable Consumption and Production (ERSCP 2019), October 15-18, Barcelona, Spain.

Arekrans, J. (2019). Report on non-technical barriers and legal issues. Public delivery (D6.1) of ReCiPSS project, https://doi.org/10.5281/ZENODO.3687542

Arekrans, J., Ritzén, S., Laurenti, R., & Sopjani, L. (2019). Analysis of innovation management issues in barriers to circular economy. Paper presented at the 19th CINet Conference: Innovating in an Era of Continuous Disruption, Odense, Denmark, September 8-10, 2019.

Hamano, M., **Arekrans**, J. & Ölundh Sandström, G. (2020). Toward circular economy implementation: A tool for integrating circularity indicators into portfolio management. Electronics Goes Green 2020+: The Story of Daisy, Alexa and Greta (online conference)., September 1, 2020.

Arekrans, J., Ritzén, S., & Laurenti, R. (2021). Circular Economy Transitions: The Maturity of Incumbents. Paper presented at the 22nd CINet Conference: Organising Innovation for a Sustainable Future, September 12-14, 2021.

Arekrans, J., Ritzén, S. (2022). Managing Circular Innovation Goals and Measurements. Paper presented at 29th IPDMC Conference: Innovation in the Era of Climate Change, Hamburg, Germany, July 17-19

Arekrans, J., Nilsson, S., Ritzén, S. (2023). Management Controls in a Circular Economy Transition. Paper presented at 30th IPDMC Conference: 30 Years of Research in Innovation and Product Development Management, Lecco, Italy, June 7-9

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1. Introduction

At the Livermore Fire Station in California, a lightbulb was installed in 1901 to provide light to firefighters. The light has been switched on for most of its life and is still in operation today, over 120 years later. While this particular bulb holds the record for the longest lifespan known, there was nothing unique about it when it was manufactured, and there are others manufactured during the same time that are still functioning. So why aren't all lightbulbs so durable? In the 1920s, technological advancements made mass production possible and caused supply to outpace demand. In 1924, finding the lightbulb market saturated and faced with declining sales, top representatives from the largest manufacturers met and founded the Phoebus cartel. This supervisory body introduced an engineered shorter lifespan for lightbulbs (1,000 hours). This reduced lifespan meant that more lightbulbs were discarded every year and thus more bulbs could be sold, increasing the throughput of bulbs in the economy (Valant, 2016). This agreement is a classic example of *planned obsolescence*, a term that stems from the essay "Ending the Depression through Planned Obsolescence" by Bernard London (1932). In essence, London suggested that the obsolescence of capital and consumer goods should be decided at the production stage in order to revitalize the economy. While obsolescence is only one out of several underlying factors contributing to unsustainable production and consumption, it exemplifies the power that large industrial firms and designers have in steering the consumption of goods through decisions in early product development.

The starting point for this research is the true and justified belief that business-as-usual following the current pattern of industrial production and consumption is unsustainable. Global greenhouse gas emissions continue to increase, with unequal historical and ongoing contributions arising from unsustainable energy use, land use and land-use change, lifestyles, and patterns of consumption and production among individuals, within countries, and across regions (IPCC, 2023). Between 1970 and 2015, global annual natural resource extraction grew from 27 billion tons to 92 billion tons, a figure that is estimated to reach 190 billion by 2060 (IRP, 2019). Even when considering natural resources that appear to be abundant, the growing world population and rising standards of living increase the cost of material extraction from the lithosphere and the consequent waste streams that must be handled (Meadows et al., 2006). At the same time, innovation in products and technology in the "take-make-waste" industrial system boosts demand for a steady supply of critical raw materials (Mathieux et al., 2018). In other words, the growth of an economic system that relies on the sales of physical goods manufactured using finite natural resources cannot be sustained indefinitely.

Large established industrial firms have a peculiar position with respect to sustainable development. They both contribute to several environmental sustainability problems through linear production and consumption, and at the same time have the resources and capabilities to promote and achieve significant changes in forms of production and consumption. Historically, businesses have viewed long-term sustainability and short-term profit maximization as trade-offs (Friedman, 1970). However, this perspective has started to

shift in recent decades (Hart, 1997). In this vein, Porter and van der Linde (1995) argued for environmental regulation as a driver to stimulate innovation. Innovation can lead to competitive advantages either directly (thanks to innovative products, services, or processes) or indirectly, through productivity improvements or by taking preemptive measures to comply with the most stringent regulation before competitors do. This argument references firms' ability to innovate and improve in response to dynamic business environments. However, in order to seize opportunities, firms face managerial challenges related to incomplete information, inertia, and control problems in aligning individual, group, and corporate motives and incentives (Porter & Van Der Linde, 1995). Therefore, firms must systematically consider their approach to integrating sustainability into management practices. Several levels of organizational activities must be taken into account, and, preferably, innovation must be facilitated and business opportunities consolidated with sustainability. Incremental approaches to integrating sustainability in business organizations may reduce both waste and costs, but these approaches do not address the root causes of sustainability issues quickly enough. This points to the need for a radical shift in production and consumption rather than an incremental one (Clarke & Roome, 1995; Kennedy et al., 2017).

The circular economy (CE) has gained significant traction among policymakers and the business community as an economic system that replaces the "end-of-life" concept with a mandate to reduce, reuse alternatively, recycle, and recover materials across the production, distribution, and consumption cycle (Kirchherr et al., 2017). A CE transition shifts the focus from only looking at downstream concepts such as end-of-life, waste management, and recycling towards keeping products, components, and materials at a high level of utility and in use. Proponents of the CE argue that it holds significant untapped business potential, effectively decoupling resource utilization from economic growth (e.g., Ellen MacArthur Foundation, 2013). While newly formed companies can reap some of this potential, the feasibility of closing material loops is highly dependent on choices made in early product development, giving existing industrial incumbents a competitive advantage in CE transitions (den Hollander et al., 2017; Lieder & Rashid, 2016).

A shift from a linear business logic cannot be achieved through stepwise incremental changes to existing products and technology alone; rather, it demands the exploration of new products, services, and business model configurations that can potentially reach new customers. However, firms tend to favor activities that perpetuate the status quo (i.e., exploitation) over searching for the business opportunities of tomorrow (i.e., exploration), as March has highlighted (1991). Although many businesses have shown interest in CE as a way to become more sustainable, we lack examples of large incumbent firms that have entirely transformed their business (Kirchherr & van Santen, 2019). To adopt CE principles, achieve radical innovation, and support a CE transformation of their business, incumbent firms need to question practices that are built upon a linear logic. Specifically, the literature has raised core questions such as organizational path dependencies, inertia, culture, and learning are topics that are crucial to better understanding how to facilitate CE

implementation and adoption at incumbent firms (Korhonen, Nuur, et al., 2018; Santa-Maria et al., 2021).

Adopting CE principles is a complex challenge for incumbent firms. When managers break down complexity to make sense of CE and guide its adoption, they must recognize and reconsider existing ways of thinking and doing, so that the organization's activities point in the right direction and do not stagnate (Kaplan, 2008; Smith & Tushman, 2005). While an increasing number of firms have described high-level strategic plans and targets that include CE (Opferkuch et al., 2021), the daily decisions made by thousands of people in an organization do not rationally consider strategic targets or plans in every situation they face. Instead, between strategy and operation, there are managerial controls at play that guide employees' decision-making and behavior (Merchant & Van der Stede, 2017). These controls can potentially help managers to both support the achievement of organizational goals and empower employees to identify opportunities and solve problems: i.e., achieving a balance between control and innovation (Ahrens & Chapman, 2004; Simons, 1995b). This makes managerial controls attractive mechanisms to consider in CE adoption. However, for large and established firms, existing controls are built on a linear logic that may be at odds with CE principles.

This thesis aims to delve into the managerial challenges that large, established industrial firms encounter when implementing CE principles and how they address them, particularly through managerial controls. Additionally, it aims to provide insights into how management can facilitate radical circular innovation and support a circular transformation of incumbent firms. This inquiry is believed to result in both practical managerial implications and contributions to the emerging body of research examining the management and adoption of CE in firms. Furthermore, it contributes to the innovation management literature by applying established concepts to a new phenomenon, adding empirically based insights.

This thesis is structured in six main sections. First, the theoretical background is presented to give the reader context to the thesis and to introduce the fundamental concepts used. Second, building on this theoretical background, the thesis's research questions are presented. Third, the research setting and research design are presented, discussing both overarching issues such as the researcher's worldview and details of research methods used in the appended papers. Fourth, there is a brief summary of the main findings of the appended papers to give an overview of how they contribute to the thesis. Then, a meta-analysis of the appended papers as a way to bridge and theorize using their specific findings, particularly the empirically based ones. Finally, the conclusions of this thesis and the implications for practice are presented.

2. Theoretical Background

This thesis explores the implementation and adoption of CE at large, established industrial firms as a way of decoupling resource consumption from economic growth. While CE principles are foundational, and their implementation in large established firms is the core

studied phenomena in this thesis, there are prerequisites to the successful broader adoption of CE in these firms. Therefore, this thesis explores effective innovation management, organizational learning, novel business models, sustainability integration, and the use of management controls. By collectively reviewing these areas, we can gain a comprehensive understanding of the challenges and opportunities that large industrial manufacturers confront when implementing CE. This integrated approach highlights knowledge gaps and underscores the essentials for propelling scholarly comprehension and real-world implementation of CE and sustainable business practices. Notably, CE implementation can be studied from a range of different perspectives; in this comprehensive summary a research journey is outlined from a macro perspective (e.g., what does a systemic CE transition entail for large incumbents?) towards a micro perspective (e.g., how is CE perceived or cognitively framed by individual managers?).

In the following sections, the theoretical background of this thesis is presented. First, the concept of CE is described and discussed in relation to sustainability, in order to give the necessary context and historical background. Second, different perspectives on innovation in relation to sustainability and CE are presented as a way to narrow the scope to business organizations. Third, the two theoretical perspectives of exploration-exploitation and management control systems are introduced, as these perspectives are used in the empirical studies to understand managerial challenges in adopting CE. Lastly, in conjunction with these two theoretical perspectives, the need to consider how different cognitive framings emerge and gain traction within organizations is discussed. The idea of different framings is used in the analysis and discussion as a way of synthesizing and theorizing the empirical findings of the appended papers.

2.1 The Circular Economy – A New Perspective on Sustainability?

Historical development and antecedents of CE

The key objective of a circular industrial economy is to keep the economic value and utility of stocks of manufactured objects and materials as high as possible, for as long as possible. Consequently, we can view growth in terms of the quality and quantity of said stocks rather than only as throughput (Stahel, 2019). The ideas that CE encompasses have many antecedents and are rooted in different schools of thought. Ghisellini et al. (2016) point to three significant sources: first, Kenneth Boulding's (1966) metaphor of the planet Earth as a spaceship—a closed system where no significant amount of matter is exchanged with the outside environment; second, general systems theory (e.g., von Bertalanffy, 1969), which stresses that every agent needs to be understood by its relationship with other actors in a system, thereby promoting systemic thinking and holism; and third, industrial ecology (Erkman, 1997), a concept where the industrial system and its environment are viewed not in a necessarily parasitic relationship but as a joined ecosystem characterized by flows of materials, energy, and information, together with the provision of resources and services from the biosphere. Other scholars (e.g. Korhonen, Honkasalo, et al., 2018) have also pointed to other antecedents that have become incorporated within the umbrella term CE, such as product-service systems (Tukker, 2004), the sharing economy, collaborative

consumption (Belk, 2014), the performance economy (Stahel, 2010), and cradle-to-cradle design (McDonough & Braungart, 2002), to only name a few. Of course, although these ideas are conceptualized and theorized in new ways, most of them are not new. Before the Industrial Revolution, people lived in scarcity and were driven by need (Stahel, 2019). There was little to no surplus of manufactured goods, and the few items that people owned also had a high utilization rate, were used for a long time, and were repaired or used for other purposes when needed. In other words, little waste was generated.

Definitions and relation to sustainability

The broad range of ideas wrapped up in the concept of CE has led to considerable confusion and a lack of consensus on its definition (see Table 1). In response, Kirchherr et al. (2017) reviewed 114 definitions of the circular economy and found that the most common definition to date was that of the Ellen MacArthur Foundation (2013, p. 7). They also found most definitions failed to address all three dimensions of sustainability. Instead, they noted that the most emphasized aim of CE was economic prosperity (incorporated into 46% of the definitions in their sample), closely followed by environmental quality (38%). Only a few definitions considered the social dimension (roughly 19%).

Table 1: Definitions of the circular economy.

Source	Definition
Ellen MacArthur Foundation 2013 (p. 7)	An industrial system that is restorative or regenerative by intention and design. It replaces the ”end-of-life” concept with restoration, shifts towards the use of renewable energy, eliminates the use of toxic chemicals, which impair reuse, and aims for the elimination of waste through the superior design of materials, products, systems, and, within this, business models.
Kirchherr et al. 2017 (p. 229)	An economic system that replaces the “end-of-life” concept with reducing, alternatively reusing, recycling, and recovering materials in production, distribution, and consumption processes. It operates at the micro level (products, companies, consumers), meso level (eco-industrial parks), and macro level (city, region, nation and beyond), with the aim of accomplishing sustainable development, thus simultaneously creating environmental quality, economic prosperity, and social equity, to the benefit of current and future generations. It is enabled by novel business models and responsible consumers.
Geissdoerfer et al. 2020 (p. 3)	An economic system in which resource input and waste, emissions, and energy leakages are minimized by cycling, extending, intensifying, and dematerializing material and energy loops. This can be achieved through digitalization, servitization, sharing solutions, long-lasting product design, maintenance, repair, reuse, remanufacturing, refurbishing, and recycling.

Several other studies have also pointed to this failure to consider social aspects (e.g. Corvellec et al., 2021; Geissdoerfer et al., 2017). Ultimately, in an effort to respond to the flaws they found in existing definitions, Kirchherr et al. (2017) suggest their own. However, even this admirable attempt has been questioned. Geissdoerfer et al. (2020) pointed out that this definition simplifies the term *end-of-life* to mean disposal and has a reduced focus on other lifecycle stages. Furthermore, they find the focus on sustainability unwarranted,

thereby putting more emphasis on defining CE as a reflection of "as-is" rather than "as it ought to be." This critique is most likely due to the previous study by Geissdoerfer et al. (2017), in which they explore the linkages between sustainability and CE. Geisdoerfer et al.'s (2020) categorization identifies eight different scholarly interpretations of this link, and their own analysis argues that CE has a subset relation to sustainability: that is, that CE is one among several solutions for fostering sustainable systems, without implying a hierarchy between other complementary strategies that managers and policymakers might adopt.

Despite the noted critique, the definition by Kirchherr et al. (2017) is considered sufficient to cover the scope of this thesis. With this definition in mind, the thesis investigates the micro level (i.e., products, companies, consumers), primarily from the perspective of large, established industrial firms. Furthermore, the operationalization of CE through R-strategies is discussed in the following section, followed by the use of business model innovation.

Operationalizing CE principles through R-strategies

The empirical part of this thesis investigates the perspective of large industrial incumbent firms. Implementing a circular economy entails significant changes for industrial companies. One apparent change concerns resource flows: i.e., using a smaller volume of resources and not letting refined resources go to waste. As a consequence, both upstream and downstream material flows must be considered in new ways, investigating which materials and which resources are used in the company's products, what happens to the resources after the product is sold, how products are used over several lives, how they are repaired and remanufactured, and finally, how the material is handled and recycled at product end-of-life. Notably, all three definitions in Table 1 above refer to different strategies or ways of retaining the value of manufactured stock and materials (reuse, recycling, recovery, repair, remanufacturing, and refurbishing). These are central elements of changes to industrial firms' products and business models. While the number of included value-retention options (i.e., R-strategies) varies, and the definitions of the R-strategies tend to vary as well (Reike et al., 2018), one frequently recurring R-strategy framework is that suggested by Potting et al. (2017) (see Figure 1).

Circular Logic	Reasoning	R-strategy	Example
	Smarter product use and manufacture	**R0 - Refuse**	Make product redundant by abandoning its function or by offering the same function with a radically different product
		R1 - Rethink	Make product use more intensive (e.g. by sharing product)
		R2 - Reduce	Increase efficiency in product manufacturing or use, consuming fewer natural resources and materials
	Extending lifespan of a product and its parts	**R3 - Reuse**	Reuse another component or discarded product which is still in good condition and fulfils its original function
		R4 - Repair	Repair and maintenance of defective product so it can be used with its original function
		R5 - Refurbish	Restore and old product and bring it up to date
		R6 - Remanufacture	Use parts of discarded product in a new product with the same function
		R7 - Repurpose	Use a discarded product or its parts in a new product with a different function
	Useful application of materials	**R8 - Recycle**	Process materials to obtain the same (high grade) or lower (low grade) quality
		R9 - Recover	Incineration of material with energy recovery
Linear Logic	*Take-make-dispose*		

Figure 1: R-strategy framework for achieving CE, after Potting et al. (2017).

Besides offering actionable examples of how each strategy can be achieved, the R-strategy framework also suggests a hierarchy among the different strategies (e.g. recovery is “less” circular than reuse). In Potting et al. they only suggest this hierarchy as a rule of thumb that has clear exceptions and that mainly refers to the underlying reasoning behind the strategies rather than an objective and measurable truth. Other scholars have proposed similar rankings: for example, Walter Stahel’s famous inertia principle (2010, p. 195):“Do not repair what is not broken, do not remanufacture something that can be repaired, do not recycle a product that can be remanufactured. Replace or treat only the smallest possible part in order to maintain the existing economic value of the technical system.”

Which R-strategies are targeted?

Reike et al.’s study (2018) found that about 60% of scholars who write about R-strategies do suggest a hierarchy among them. However, given the differences in approaches to R-strategies, it is challenging to define which is "more circular." Therefore, CE researchers have engaged in considerable discussion on how to measure circularity as a way to monitor and guide implementation on various levels. Saidani et al. (2019) reviewed 55 sets of circularity indicators from the academic and grey literature. They argued that indicators can serve various purposes in CE transitions. In business organizations, they can serve as key performance indicators to monitor the implementation of CE practices. Furthermore, they help support a common language to operationalize and spur knowledge dissemination regarding the confusing concept of CE. Lastly, indicators can provide a basis for decision-making without conducting a complete lifecycle analysis, which can be time-consuming. However, on this micro level of analysis, researchers have found that the indicators used to date have significant flaws. For example, 90% included recycling loops but less than half (45%) considered all possible R-strategies. This is arguably problematic, as it may dilute the concept of CE to recycling, even though this is the strategy with the least circularity potential (Potting et al., 2017; Reike et al., 2018).

Pauliuk (2018) points out that CE is still in the early stages of implementation among industrial firms and that linkages to sustainability, social risks, and ethical responsibility are still under debate. These findings are echoed by Opferkuch et al.’s more recent content analysis of sustainability reports (2021), which found that most frequent discourse only considered the environmental dimension of CE, specifically regarding waste management operations or resource management. The authors note that this points to a relatively incremental and linear transformation of industrial firms rather than a radical one.

Evidently, there are an increasing number of incumbent firms implementing CE. However, there is a disparity between their high ambitions and actual achievement in terms of radical changes that truly move beyond linear production and consumption. This underscores the need to consider how radical circular innovation can be facilitated in industrial firms.

2.2 Innovation as a Vehicle to Implement Sustainability and CE

What is innovation?

Although innovation can mean many different things in various contexts, at its essence, we can state that innovation entails some form of change. The distinction between radical and incremental innovation (a difference of magnitude) often relates to the extent to which the innovation departs from existing practice (Dewar & Dutton, 1986). Incremental innovation thus includes minor improvements and simple adjustments, often to existing technology. In contrast, radical innovation encompasses fundamental changes that clearly depart from existing practices and embody new knowledge. Frequently, radical innovation is associated with business model innovation, although product-level innovation can be equally radical (Crossan & Apaydin, 2010). However, this way of distinguishing between radical and incremental depends on the unit of adoption: i.e., the innovation is new to whom? Song & Montoya-Weiss (1998) point out that novelty is most often viewed relative to the firm or the outside world, and the predominant measurement often relies on managers' perception from the production side, rather than customers' perceptions on the consumption side. Radicalness can also be defined by the degree of risk it entails, as it is often considered to be riskier than more minor improvements. Still, the level of risk is often also based on individual judgment, experience, and perspective. Altogether, the distinction is not clear-cut or objective, making it challenging to define or measure radicalness, and the fact that radical vs. incremental is a continuum is often ignored and instead viewed as a dichotomy, meaning polar opposites (Dewar & Dutton, 1986).

Innovation for sustainability

Profit maximization traditionally has dominated much of the management literature as the sole objective of business organizations (see e.g. Friedman, 1970). Therefore, sustainability (particularly its environmental dimension) has historically been viewed as a cost of doing business framed in terms of a trade-off between ecology and economy (Stonehouse & Snowdon, 2007). Despite scholarly developments in operationalizing sustainability in management practices (for example, through corporate social responsibility initiatives or corporate sustainability [Dahlsrud, 2008]), in practice these initiatives have been criticized as being highly fragmented and lacking strong links to core business strategies (Porter & Kramer, 2017), and as not integrated with traditional managerial control systems to balance conventional economic objectives (Crutzen et al., 2017; Gond et al., 2012). However, one stream of scholars and practitioners has documented a shift in businesses' rationales for sustainability implementation from concerns such as pollution control and waste reduction towards seeing it as a source for revenue growth and competitive advantages (Hart, 1997; Porter & Van Der Linde, 1995). This newer perspective closely couples sustainability with innovation in products and services, a trend captured in the literature on sustainability-oriented innovation (Hansen & Große-Dunker, 2013).

While incremental approaches such as "end-of-pipe" technical additions or eco-efficient optimizations of current business activities may help firms reduce waste and eliminate unnecessary costs, such changes do not address the root causes of sustainability issues, a

goal that calls for more radical shifts (Clarke & Roome, 1995; Kennedy et al., 2017). Consequently, some companies are now broadening their views to look at the socio-economic-technical-ecological systems in which they operate rather than simply considering themselves as providers of products (Roome & Louche, 2015). However, alongside the challenges of innovation management in general, there are specific issues relating to sustainability to be addressed throughout innovation processes aimed at improving sustainability. For instance, there is a high degree of complexity involved in reconciling potential conflicts between sustainability goals and stakeholder demands. Sustainability targets can also be ambiguous due to scientific uncertainty, and more sustainable technologies and offerings can be competence destroying innovation for incumbent firms (Hall & Vredenburg, 2003).

Adams et al. (2016) review the extant literature on firm-level sustainability-oriented innovation and synthesize their findings into a framework that commences with firm-level incremental changes in response to regulatory stimuli and culminates with radical systems-level changes (see Figure 2). The framework captures the focus of innovations (*technology ↔ people*), the firm's view of itself in relation to society (*insular ↔ systematic*), and the extent to which innovation extends across the firm (*stand-alone ↔ integrated*). Furthermore, they outline three different firm-level approaches to sustainability-oriented innovation: operational optimization (i.e., eco-efficiency), organizational transformation (i.e., new market opportunities), and systems building (i.e., societal change).

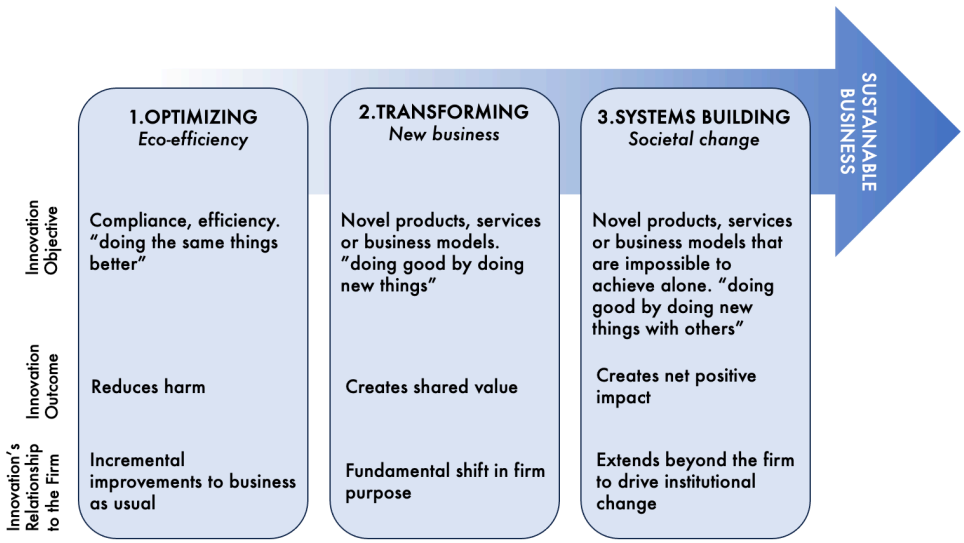


Figure 2: Firm-level sustainability-oriented innovation, after Adams et al. (2016).

This framework is well grounded and captures significant developments in theory. However, the authors note that the current literature is in the early stages of theory building: i.e., the "what" phase. In addition, the level of analysis concerns firm-level responses. Therefore, they suggest that future research should focus on *how* to transition between the approaches: for example, understanding the knowledge, resources, competencies, and capabilities that

are required within firms to facilitate managerial action capable of transforming firms towards greater sustainability (2016).

Innovation for circular economy – the role of business model innovation

Much of the CE literature is clear on the need for innovation in product design (den Hollander et al., 2017), product-service systems (Tukker, 2015), and enabling technology (Asif et al., 2018). However, one core problem for incumbent business organizations is the fact that conventional business models do not offer much potential for closing material loops (Rashid et al., 2013). Consequently, CE requires the design and implementation of innovative business models that aim to minimize resource utilization, for as long as possible, while maximizing the value extracted in the process (Geissdoerfer et al., 2020). This new need has led to a body of literature that bridges CE and business models, building on sustainable business models (Boons & Lüdeke-Freund, 2013; Schaltegger et al., 2012, 2016).

The literature on business model innovation for CE has two distinct types of work, either the prescriptive and conceptual, or, as a process and strategic leverage. The former approach aims to describe characteristics and frameworks for how circular business models should be developed (Bocken & Ritala, 2021; Lewandowski, 2016). However, the relative newness of CE implementation in firms means that much of this descriptive literature stream is nevertheless mostly conceptual or examines individual case studies. The second stream in the literature pays less attention to specific business model designs and considers the business model innovation process set in an organizational context (Diaz Lopez et al., 2019; Frishammar & Parida, 2019). Specifically, business model innovation can be viewed as a vehicle for businesses to systematically implement CE at the organizational level, effectively challenging their core business logics and addressing the alignment of incentives among different stakeholder groups (Geissdoerfer et al., 2020). The for this latter perspective is especially clear when implementing R-strategies with a high level of circularity, as this type of implementation goes beyond incremental changes and contributes to more systemic and radical transformation (Nußholz, 2017).

Notably, however, while several scholars emphasize the need for firms to make strategic choices when adopting CE, business model innovation is often viewed as an isolated process separate from strategy. In addition, while several tools and methods are being developed for business model innovation, in their literature review, Pieroni et al. (2019) warn that failure to consider the organizational context and integrate aspects such as normative management and change management with those tools and methods could hinder their practical applicability. Similarly, in a recent review and multiple-case study, Santa-Maria et al. (2021) found that issues such as change management and organizational inertia remain underexplored when addressing the complex challenge of innovation toward sustainable and circular business models. A failure to consider these aspects may be especially detrimental to established firms. In particular, Briguglio et al. (2021) distinguish between existing incumbent firms and those that are “born circular” and find that the latter type of businesses generally face fewer barriers to CE implementation, and the barriers they do face are different from those that incumbent firms face. Similarly, when examining barriers to

circular business model innovation, Guldmann and Huulgaard (2020) also found that large companies faced more internal barriers to innovation related to organizational inertia arising from path dependency and lock-ins than do small companies. Focusing on incumbent firms, Hofmann and Jaeger-Erben (2020) found the adoption of new sets of performance measurements as an intervening mechanism in the organizational transition toward CE. Interestingly, they note that the adoption of new performance indicators is potentially hampered by senior employees who have made their careers in the company based on old linear logics and understandings of performance. Combined, these studies raise questions about organizational learning and the compatibility between the CE transformation and existing managerial control systems at incumbent firms, topics which will be discussed in the following sections.

Innovation in this thesis

In this thesis, *radical circular innovation* refers to significant departures from existing practices in production and consumption with the aim of replacing the concept of end-of-life for products, components, and materials and instead seeks options that retain the most value for as long as possible, bundled into products, services, and business model configurations. While this thesis pays particular attention to radical circular innovation, it is not a radical products, services, processes, or business models as an outcome *per se* that is of primary interest for research purposes. Instead, the interest lies in discovering *how to facilitate the process* for radical circular innovation in order to support CE adoption at large industrial firms and lead such firms towards a CE transformation. In addition, radical innovation is not studied in isolation. Instead, it is necessary to compare and contrast it with ways in which incremental innovation is facilitated to provide greater context.

With this said, it should be noted that the *business model* innovation literature is considered the most relevant stream of CE literature for this thesis connecting CE and innovation, for three primary reasons. First, it holistically considers several interacting elements of business activities (i.e. reaching customers, creating new revenue streams, changing value propositions, etc.), which arguably may be particularly relevant for the systemic approach that CE implementation calls for, especially with respect to more radical R-strategies (Nußholz, 2017; Potting et al., 2017). Second, as highlighted above, one stream of the business model literature (e.g., Schaltegger et al., 2012; Stubbs & Cocklin, 2008) takes organizational context into consideration (i.e., interactions between business models, strategies, and organizational transformation)—a perspective which may be imperative for making enduring radical changes. Third, with the thesis being pragmatically guided by industry needs, several large established industrial companies are less experienced in working with business model innovation versus product innovation, and managers view it as more radical (Crossan & Apaydin, 2010; Song & Montoya-Weiss, 1998).

Noteworthy, as both CE and innovation are complex topics of which different individuals and organizations have their own understandings, there is therefore an inherent need to approach these concepts from multiple perspectives in order to gain a more comprehensive understanding of what is happening in practice. In other words, the design of this research

was not limited to, for example, the definition of radical circular innovation provided above, or to investigating only product-level innovation as an outcome or portfolio management as a determinant of such innovation (Crossan & Apaydin, 2010). Instead, an inclusive and pragmatic approach has been chosen, which will be discussed in the section on research design.

Next, the two primary theoretical concepts used during the empirical research will be introduced: exploration-exploitation (March, 1991), and management control systems (Merchant & Van der Stede, 2017). These are selected for their usefulness in addressing many of the organizational challenges discussed above, as will elaborated on in the coming sections.

2.3 Organizing and Managing Innovation

Exploration and exploitation and organizational learning

Organizational behavior and decision-making are, to a large extent, guided by learned experiences. Thus, instead of looking at innovation as bundled into products, services, processes, or business models, we can distinguish the knowledge creation and organizational learning that innovation entails. March's seminal work (1991) introduced the concepts of exploration and exploitation in organizations. Here, exploitation encompasses knowledge-intensive activities such as searching, creating variations, risk-taking, experimentation, being flexible, playing, and discovering. In contrast, exploitation activities involve refinement, choice, selection, production, efficiency, execution, and implementation. However, the challenge that March (1991) highlights is that the two sets of activities significantly differ in their expected values, returns, timing, and distribution within and beyond organizational boundaries. Consequently, he argues the allocation of resources to either type of activity is significantly dependent on comparisons made between the two. In turn, these comparisons may be influenced by a range of divergent strategies, goals, and social contexts and are subject to personal preferences, beliefs, experiences, and levels of risk tolerance. Thus, at large, established organizations, the advantages that favor exploitation accumulate and prevail at the cost of exploration. For a business to remain relevant and profitable in both the short and long terms, exploration and exploitation must exist in an adequate balance (He & Wong, 2004; Tushman & O'Reilly, 1996), and achieving this balance helps firms avoid the various traps that lead to overemphasis on exploitation (Ahuja & Morris Lampert, 2001; Levinthal & March, 1993).

Can we achieve both exploration and exploitation?

Levinthal and March (1993) problematized the processes of organizational learning with regard to exploration and exploitation. They argue that the central problem for exploration is that fast-learning and successful firms tend to be very responsive to the clear and early feedback that exploitation activities generate, leading them to shift resources away from exploration activities that they associate with uncertainty and failure. Consequently, they argue that sustaining exploration requires changes in incentives, organizational structures, individual beliefs, and selection processes. The literature on ambidextrous organizations has

described different approaches of organizing to balance balancing exploration and exploitation. Tushman and O'Reilly (1996) define ambidexterity as an organization's ability to implement both incremental and revolutionary change. Many of the different approaches can be captured through the balancing modes of contextual ambidexterity, organizational separation, temporal separation, and domain separation. Interestingly, Lavie et al. (2010) point towards different organizational levels, mechanisms, management roles, and challenges to each approach.

Although research on ambidextrous organizations has flourished, O'Reilly and Tushman (2013) point out a lack of qualitative insight into how business leaders manage the interfaces between exploration and exploitation and the inevitable conflicts that arise between them. In light of this, Hansen et al. (2018) investigated failed radical "green" technology innovation projects at SMEs from the ambidexterity perspective. They argue that a core challenge for ambidextrous businesses is the extent, type and timing of separation and integration between exploration and exploitation. They found that failing innovation projects suffered from separational drift: i.e. the separation of exploration and exploitation was prematurely eroded before the outcome was ready for operationalization, with exploitative routines "leaking in" to explorational activities. Moreover, they found that exploration needed to be "cross-functional" in order to pursue market innovation and technological innovation simultaneously, and that R&D programs that focused on technological innovation alone were less likely to succeed. Notably, they found that cross-functional approaches to exploration were particularly relevant yet challenging for green technology innovation, due to the significant complexity and uncertainty resulting from factors outside the firm. These findings are interesting to highlight in relation to the complexity of CE and of large industrial firms, where exploitative behaviors may become routinized and coordination efforts more formalized in comparison to SMEs.

Raisch and Birkinshaw (2008) found that we know relatively little about individual capacities to pursue exploration and exploitation concurrently (i.e., contextual ambidexterity). Instead of, for example, separating exploration and exploitation through organizational structures, contextual ambidexterity utilizes behavioral and social means to integrate exploitation and exploration (Andriopoulos & Lewis, 2009). Therefore, successfully balancing exploration and exploitation and implementing radical innovation within the main business line requires middle management to hold an ambidextrous mindset (Gassmann et al., 2012) and for leaders to provide supporting infrastructure, processes, and systems within organizational units that enable individuals to tackle contradictions (Gibson & Birkinshaw, 2004; Lavie et al., 2010). However, accomplishing all this faces significant challenges. As March (1991) notes, there are cognitive limits that individuals encounter when trying to simultaneously embrace both exploration and exploitation, which creates tensions between long-term and short-term ambitions.

To zoom in and explore the individual level and the role of prior knowledge (Cohen & Levinthal, 1990), the concept of cognitive frames (Walsh, 1995; Weick et al., 2005) is introduced in the following section. Then, zooming back out to explore how managers create

the supporting infrastructure, processes, and systems, the concept of management control systems (Merchant & Van der Stede, 2017) is introduced thereafter.

Cognitive frames and sensemaking processes

To describe how individual managers interpret situations, the idea of *cognitive framing* is recurring in literature (Walsh, 1995). This concept builds on bounded rationality: i.e. the stance that people cannot make rational decisions that consider all available information. In management research, cognitive framing has been used in studying firms and managers who face strategic decisions and need to interpret an abundance of information. More specifically, researchers have argued that managerial cognitive frames drive organizational action by directing attention to particular issues (Smith & Tushman, 2005). Managers develop subjective representations of the information environment (in this thesis, one such representation is *the circular economy transition*), and these representations are what drive the subsequent actions that managers take (i.e. *consequences*) (Nadkarni & Barr, 2008). Cognitive frames (also referred to as frames of reference) are used to reduce the amount of information and its inherent complexity and can be described on the individual, team, or organizational levels (Walsh, 1995; see Figure 3 for an illustration).

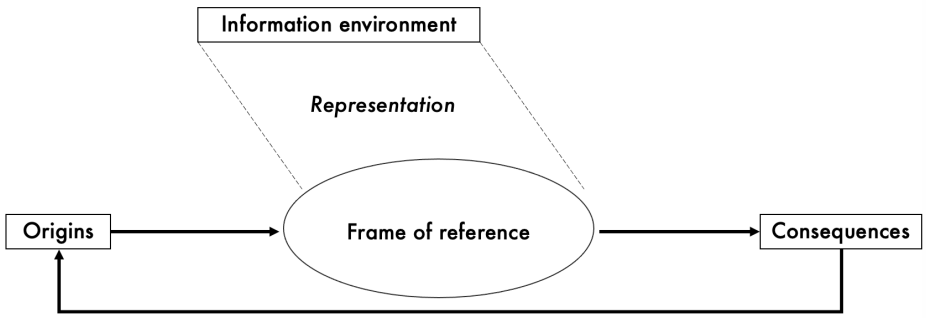


Figure 3: Illustration of cognitive frames, after Walsh (1995).

While these frames help managers to select, organize, and interpret information or a phenomenon, they can also lead to fallacies. For example, cognitive frames are built on experiences, actions, and knowledge acquired in the past (Origins), and therefore the frames reinforce themselves. A framing may, therefore, result in a limited understanding of a situation or confirmation bias (Levinthal & March, 1993) if it leads people to disregard information that does not fit their current frame of reference (Barnes, 1984). This fallacy is important to consider at an organizational level at incumbent firms, as their historical experiences and actions are learned and routinized (Cohen & Levinthal, 1990; Levitt & March, 1988). Because framings may differ between individuals and organizational subunits, framing "contests" can occur (Kaplan, 2008), i.e. sensemaking processes through which individuals engage other actors to build legitimacy and interest for their particular framing within the organization. When successful, *dominant* collective frames can emerge that guide the firm's subsequent actions.

Cognitive frames: Innovation and sustainability

Bessant et al. (2014) find that framing has been conceptualized and studied in many different forms with respect to innovation: for example, through descriptions of "interpretive schemas" (Greenwood & Hinings, 1993), "recipes" that managers share (Spender, 1996), "dominant logics" (Prahalad, 2004), "cognitive rules' (Geels, 2004) or "cognitive boxes" (Bessant et al., 2014), all of which shape decision-making and action. However, studies of CE adoption have not used the concept of cognitive framing extensively. Bertassini et al.'s review of the literature (2021) found that the limited amount of CE literature regarding organizational behavior has been oriented towards the ways that mindsets, capabilities, and competencies function as barriers to CE implementation, and not how they might serve as mechanisms for enabling CE transformation. Nevertheless, a growing body of literature has looked at corporate sustainability through the lens of cognitive frames (Dyllick & Hockerts, 2017; Hahn et al., 2014; Hockerts, 2015). Much of this research focuses on the tensions between decisions that support sustainability versus those that support business-as-usual, and has argued for new managerial approaches that do not consider tensions as either/or decisions or trade-offs but rather as necessary paradoxes that should be "worked through" in a sensemaking process (Luscher & Lewis, 2008). One recent example is Haffar and Searcy (2019), who investigate how dominant logic influences decision-making regarding sustainability tensions; they show how different logics can lead towards either short-term reductionist behaviors and decisions, or, to a more holistic consideration of long-term sustainability integrated with business strategy. Notably, they found that the group whose behavior was reductionist tended to fail to integrate sustainability considerations into the company mission and values, decision-making routines, and information systems, for example. This points to a relationship between managerial control systems and dominant logics concerning sustainability.

To conclude, organizing for radical innovation in order to tackle sustainability issues faces challenges at large incumbent firms. Specifically, when contrasted with incremental innovation and business-as-usual, the complex decision-making involved and tensions among different goals and requirements may lead to avoidance and reductionism, steering resources away from exploration activities that are associated with uncertainty and risk-taking. This tendency is influenced by a range of divergent strategies and goals, as well as the social context, and is subject to personal preferences, beliefs, and experiences (i.e. cognitive framings). In the literature on cognitive framing, analyses of mechanisms that influence how dominant framings come to prevail are nascent. Consequently, as research on contextual ambidexterity illustrates, managers need to build processes and systems that enable and encourage individuals to make their own assessments of organizational tensions and conflicting goals. To that end, this thesis adds the perspective of management control systems as mechanisms to influence the dominant framing of CE and overcome organizational barriers faced in CE adoption. In particular, this is done by studying how the management control systems can support or impede radical circular innovation in large industrial incumbent firms.

2.4 Management Control Systems: Efficiency vs Innovation

What are managerial controls?

Much of the literature on management control stems from accounting and has traditionally emphasized the use of formal and often measurable financial data in managerial decisions. This classic perspective perceives Management Control Systems (MCS) as tools for executing and monitoring predetermined objectives, aiming to minimize variance and foster organizational efficiency. However, this perspective has evolved to incorporate a broader range of information (Chenhall, 2003). As Anthony and Govindarajan (2006) note, the scope of management control systems now encompasses a wide range of activities, such as organizational planning, coordinating organizational units, communicating and evaluating information, determining necessary actions, and motivating individuals to modify their behavior in line with strategic and operational objectives. Consequently, MCSs influence employees' behaviors in desirable ways and, consequently, increase the probability that the organization will achieve its goals (Merchant & Van der Stede, 2017).

There are also several ways of categorizing different types of management controls. One approach, which is relatively straightforward since it covers many different elements in only three categories, is that of Merchant and van der Stede (2017), who distinguish between results controls, action controls, and personnel/cultural controls. They also emphasize that there are several conditions that limit the suitability of each type of management control (see Table 2 below).

Table 2: Categorization of management controls and their limiting conditions (Merchant and van der Stede, 2017).

	Result	Personnel/Cultural	Action
Description	Defining desirable results, monitoring performance towards set targets, and incentivizing success.	Employees' natural tendencies to control and/or motivate themselves, and a form of group pressure on individuals who deviate from group norms and values.	Ensuring that employees perform (do not perform) certain actions known to be beneficial (harmful) to the organization.
Limiting conditions	Knowledge of desired results Ability to influence desired results Ability to measure controllable results effectively.	Level of emotional ties between employees Employee motivation to do well.	Knowledge of desirable actions Ability to ensure that desired actions are taken.

Balancing the management control system

Arguably, there are two critical roles of MCSs: controlling the achievement of organizational goals and empowering employees to identify opportunities and solve problems: i.e., the balance between control and innovation (Ahrens & Chapman, 2004; Simons, 1995b). This balancing act bears a strong relationship to the overall business strategy of the organization or subunit. For example, businesses aiming for cost leadership or defending existing

businesses tend to control employee behaviors through relatively tight, formal financial controls and standardized operating procedures designed to maximize efficiency. In contrast, businesses competing on differentiation or those searching for new markets tend to use informal control systems, participative decision-making, and rewards for employees and managers based on forward-looking, nonfinancial performance indicators (Merchant & Van der Stede, 2017). Thus, focusing too much on the notion of *control* has been associated with narrow-focused monitoring of financial values (Ruiter et al., 2022) or tight, standardized operating procedures (Eisenhardt, 1985) and has been argued to be detrimental to innovation (Langfield-Smith, 1997; Müller-Stewens et al., 2020) which necessarily entails significant uncertainty and risk-taking, as outlined in previous sections.

The special case of result controls and innovation

For this reason, it may be particularly relevant to understand the use of results controls in the face of uncertainty and risk. This issue has been addressed in several ways: for example, in Cyert and March's seminal work "A Behavioral Theory of the Firm" (BTF) (1992), which scrutinized the decision-making process of large established firms under circumstances of uncertainty. One significant aspect of BTF concerns organizational goals as a form of managerial control, where price, output, and sales strategy are governed by five different sets of goals: production, inventory, sales, market share, and profit. As decisions are made in different parts of the organization and goals are established, adapted, and monitored in various ways, discrepancies among goals come into play and may lead to conflict. These conflicts can be resolved using "acceptable-level" rules for decision-making (i.e., centrally deciding boundaries for what is "good enough"), attending to different goals at different times, and local rationality. The local rationality of goals means that the overarching goal of an organization is broken down into subunit goals: for example, the sales department is responsible for its sales strategy and subsequent goals, whereas the production unit has its own production goals and processes. In other words, these subunit goals have a local internal logic according to their specific context and time; however, this makes it challenging for large organizations to align the goals across subunits and over time.

Such misalignment can have detrimental effects in practice. For example, Ethiraj and Levinthal (2009) found that the pursuit of multiple goals does lead to a lock-in effect for status quo. This lock-in effect derived specifically from interdependencies and trade-offs among different goals: i.e., managers knew that improving performance on one criterion could be detrimental to improvement on others and were therefore reluctant to act altogether. Such questions have also been raised as a critique of the concept of stakeholder value. Specifically, Jensen (2002) soundly refuted the idea of firms holding multiple objectives and strategies altogether, since multiple objectives create confusion and a lack of alignment, and make it impossible for managers to make rational decisions. Jensen is concerned with the pursuit of shareholder and stakeholder value, making clear that integrating sustainability into management practices produces considerable tensions that must be managed. Ethiraj and Levinthal (2009) suggest strategies such as focusing on a subset of goals, dispersing goals over different organizational units, or focusing on one goal at a time as ways to manage multiple goals. However, such strategies create challenges in

terms of incentives and coordination, which can be considered in relation to the concepts discussed in the previous section on ambidextrous organizations (Lavie et al., 2010).

Results controls in the form of performance goals may also be detrimental to exploration and learning. McGrath (2001) investigated managerial oversight: i.e. (1) setting performance goals, and (2) the amount of control over operational decisions. While it is intuitive that unambiguous goals can help absorb uncertainty, motivate teams, and manage trade-offs in decision-making, the same goals can also narrow search boundaries (relating to *exploration* outlined in the previous section) and determine how new information should be interpreted (relating to *framing*). When firms engage in exploration, where little previous knowledge is available, premature goalsetting can be problematic, as this creates boundaries regarding what is desirable performance and what is a failure. McGrath points to previous findings (e.g., those of Eisenhardt & Tabrizi [1995] and Pisano [1994]), in that when more existing organizational knowledge is applied to the problems in a project (i.e. exploitation), a management approach for narrowing and focusing the project group was associated with learning effectiveness. However, when faced with a strong emphasis on exploration, the opposite appeared to hold true. In line with this, Chiesa et al. (2009) explored how management controls were utilized in different innovation projects, specifically comparing radical innovation projects to incremental ones. A higher level of uncertainty characterized the studied radical innovation projects throughout their entire duration. This was especially true in the early stages, when technical issues, market potential, and business impact were difficult to anticipate. Consequently, they found that in order to enable organizational learning, managers relied extensively on interactive controls (e.g., senior managers actively engaging in decision-making [Simons, 1995b]) when there were many uncertainties. In addition, the use of controls was not static but evolved over time as uncertainty diminished, shifting towards more diagnostic controls of results (Chiesa et al., 2009).

Use and evolution of MCS: Potential and limitations of existing MCS frameworks

Management control systems can be studied from various perspectives, including design, implementation, use, and evolution (Hansen & Schaltegger, 2016). Traditionally, however, much of MCS research has conceptually focused on the intended design (often conceptual) or on single controls in isolation, overlooking real-world systemic use and evolution in organizational contexts (Langfield-Smith, 1997; Stringer, 2007). Instead, the evolutionary perspective acknowledges the iterative nature and learning processes associated with using controls, including sensemaking (Sundin et al., 2010). In this vein, Hansen and Schaltegger (2016) point towards the need to further understand the role of managerial controls for concerns such as developing a shared understanding of what sustainability means for companies and their stakeholders, as well as what measures and processes best enhance sustainability management and organizational learning processes. In a review of emerging themes in the management control literature, Berry et al. (2009) noted a failure to consider theoretical perspectives on sustainability and the impact of finite resources. They indicate that the overarching frameworks used in MCS have intrinsic limitations that need to be better understood in light of new business contexts and environments. Going forward, they

suggest close collaboration between researchers and practitioners to identify the trade-offs that need to be considered when designing management control systems.

In sum, using MCS to steer large organizations should not be underestimated. However, this also means that when organizations make significant strategic or organizational changes, existing managerial controls may become less effective or impede those efforts at change. This effect may be particularly true for large, established firms whose MCS has developed over time and, as noted, is frequently formalized in ways that promote efficiency rather than facilitate innovation. Therefore, when implementing CE, which arguably needs innovation and entails a fundamental questioning of the logic of linear production and consumption, a better understanding of the appropriate use of management controls is needed.

2.5 Synthesizing the Theoretical Framework

In order to summarize and illustrate the theoretical concepts in the present thesis, a tentative framework is presented in Figure 4. Notably, this figure illustrates how the theoretical concepts are studied here on both the firm level and the individual level. It also shows how management control systems and the exploration/exploitation distinction are the main factors for CE adoption this thesis is concerned with. In addition, the figure outlines two clusters of theoretical concepts used in the thesis: (1) Managing (i.e. CE implementation activities and achieving large-scale organizational change), and (2) Learning (i.e. adapting to and experimenting with new logics of production and consumption).

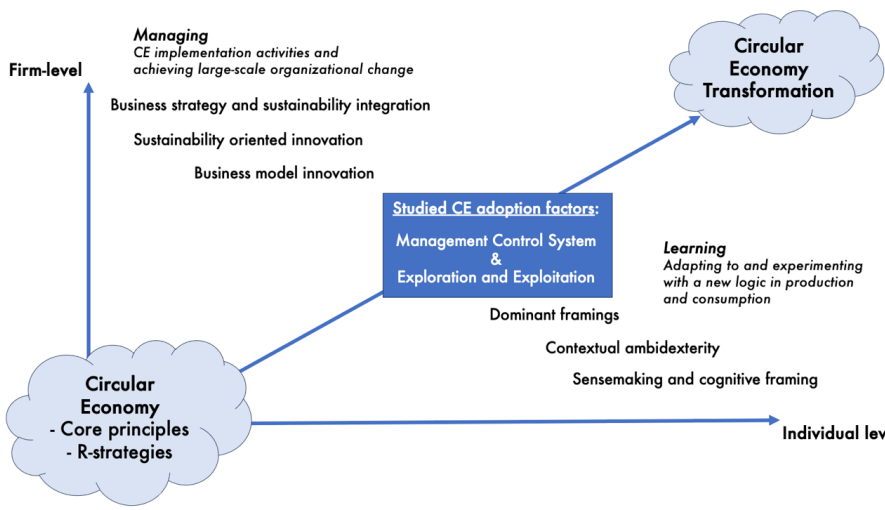


Figure 4: Representation of the framework of theoretical concepts used in the thesis, distinguishing between the individual and firm level.

In this thesis, several related terms and concepts are used, partly illustrated in the Figure 4 and reflected in the appended papers. Some essential concept pairs are *implementation and*

adoption, principles and practices, and *transition and transformation*. The author generally refers to *implementation* as the intentional installation of something novel, while *adoption* includes the acceptance of such an implementation. Similarly, *principles* can refer to abstract ideas or concepts, whereas *practices* are the useful applications of these concepts. The words *transition* and *transformation* are often used as well. *Transition* either refers to large-scale societal changes in light of sustainability issues, e.g. a CE transition or sustainability transition, or it can also relate to the shifting from one state to another, and then often used interchangeably with *transformation*, i.e. one from one state where no CE principles are implemented, towards one state where CE principles are fully adopted.

With the theoretical concepts outlined and some key terms clarified, the research questions of the thesis will be described in the following section.

3. Research Questions

The aim of CE is to transform our economic system from one based on inexhaustible resources that does not consider the implications of waste disposal towards one that acknowledges the natural limits of our resources and questions the idea of waste. This is a new logic in the business environment in which industrial firms operate and will require substantial changes. Given the pressing issues of achieving sustainability, coupled with the current linear production and consumption system, one could question whether incremental innovation alone is sufficient to address these issues with sufficient speed and decisiveness to have an impact. Accomplishing CE requires significant departures from existing practices, a fact that is particularly challenging for large, established firms, thriving with a linear logic.

Therefore, this thesis is focused on understanding how industrial firms deal with what in this thesis is referred to as *radical circular innovation*: significant departures from existing practices in production and consumption that aim to replace the concept of *end-of-life* for products, components and materials with the goal of achieving the greatest possible retention of their value over time through innovations that are bundled into products, services, and business model configurations. Hence, the first research question is:

> **RQ1.** What challenges does radical circular innovation pose for large, established industrial firms?

A shift from a linear business logic cannot be achieved through stepwise incremental changes to existing products and technology alone; rather, it demands exploring new products, services, and business model configurations, and potentially reaching new customers, which sets up requirements for how this innovation is managed. However, we know little about how incumbent firms implement CE principles or whether they implement them in ways that facilitate radical innovation. Consequently, the second research question asked is:

> **RQ2.** What aspects of managing innovation are critical for achieving radical circular innovation at large, established industrial firms?

Effective implementation and adoption of CE principles requires managers to operationalize high-level CE goals and strategies and to systematically influence daily decision-making in fundamental and coordinated ways. Management controls are mechanisms that have the potential to do this and could therefore be crucial in facilitating CE transformation at incumbent firms. However, at large, established firms, the history of the controls already in place is built upon a linear logic. Consequently, the third research question asked is:

> **RQ3.** How do management controls facilitate or impede circular economy transformation at large, established industrial firms?

The appended papers approach these questions from different perspectives and act as building blocks in achieving the overarching aim of the thesis, as will be discussed in the following sections.

4. Research Setting and Design

This section focuses on the overall research design and discusses the methods used in the appended papers. Additional reflections concerning the research design and ethical considerations are found in the discussion section.

4.1 Research Setting

Implementing CE in incumbent firms is a relatively new, complex, and evolving phenomenon. Therefore, this sets requirements regarding the type of research applicable to studying the phenomena and giving relevant and timely advice to practitioners. Research on a phenomenon such as this requires a close relationship between the researcher and the object of study (i.e., the case company) rather than the passive role of external observer. Scientists from the field of management studies can provide society or organizations with expert advice through collaborations between academia and practitioners (Committee on Science Engineering and Public Policy, 2009).

This study was conducted as a full-time PhD project with sponsorship from the EU Horizon 2020 project ReCiPSS (grant 776577-2). Throughout the investigation, I participated as a researcher in three additional projects funded by Vinnova (grants 2018–04691, 2020-00713, and 2021-03230). All projects were conducted in close collaboration with the industrial partners, who were primarily interested in the managerial implications and exchange of knowledge both with academics and with other practitioners interested in CE.

4.2 Industrial Setting

The focus of this research is large, established industrial firms. The interest in *large* firms ultimately arose from the complexity and inertia that grow along with the scale of operations

and resources, which creates a particular need for coordination and alignment at such organizations. *Established* refers to companies with a longer history and often a successful position in their target markets. In other words, these companies have established routines, processes, habits, and unspoken ways of doing what they always have done to be successful. Terms such as *manufacturing company*, *incumbent, industrial company*, *producer*, and *industrial firm* appear interchangeably in the appended papers. Unless defined otherwise, these terms refer to companies that are engaged in the production of physical goods (products or components). While the studied companies offer significantly different products and services, they have these highlighted commonalities of size, history, and production of physical goods. Furthermore, most of the companies that were involved in this research are headquartered or have operations in Sweden. This context is relevant, as the European Commission has taken considerable steps towards establishing policies for CE, which influences the studied cases due to legislative shifts.

4.3 Worldview

Epistemology roughly refers to a theory of knowledge, and ultimately asks *what* knowledge is and *how* researchers can create new knowledge. Creswell (2014) uses the more straightforward and descriptive phrase *worldview*, meaning a basic set of beliefs that guide action. Researchers' worldviews are important to consider when considering research design, as they have a significant impact on the assumptions they bring to their studies and what knowledge they believe they can create.

This thesis takes a pragmatist approach that is less dependent on philosophical worldviews and instead holds that "truth" is whatever works at the time and has useful implications for real-world problems (Creswell, 2014), such as sustainability issues (Baker & Schaltegger, 2015) or managerial challenges following the implementation of CE at incumbent firms. Pragmatism does not value an ideological stance on resolute epistemological principles but instead embraces an approach of pluralism, allowing for the integration and appreciation of multiple perspectives (York, 2009), a stance that fits well with the interdisciplinary nature of CE research. However, pragmatist researchers acknowledge that their research occurs within social, historical, and political contexts. Furthermore, in terms of what "reality" is, pragmatists believe that there is no way of truly distinguishing between what is a conception and what is objective truth; instead, what is useful is true (Patton, 2002, p. 578). In other words, the real or natural world may very well be "out there," but this is not objective. Instead, pragmatism acknowledges that while the real world exists, conceptions of it may differ in ways that often relate to the use of language.

Whenever someone formulates sentences to describe an event or fact, the views that emerge are embedded in human experience, language, and culture; this observation is in agreement with the *interpretivism* (Wicks & Freeman, 2008). For pragmatists to understand how truth serves to deal with real-world problems, they necessarily must explore the practical relevance of ideas for individuals regarding their subjective view of reality and their objectives. How individuals cognitively frame situations and come to know things as true or meaningful is captured in the concept of sensemaking (Weick et al., 2005). Studying

sensemaking and cognitive framing is thus highly relevant in a pragmatic research approach, as these processes capture how meanings that inform and constrain identity and action are materialized (Baker & Schaltegger, 2015).

Pragmatists are open to choosing certain methods, explanations, stories, and theories that produce useful results and dismissing others that are less useful (Cherryholmes, 1992). This stance opens up the possibility of using a wide variety of research methods to study the consequences of real-world actions, including qualitative, quantitative, and mixed methods (Creswell, 2014). From a pragmatist standpoint, the quality of a research inquiry does not lie in its epistemological foundation but in judging the intended purposes, available resources, procedures followed, and results obtained (Patton, 2002, p. 71).

The pragmatist approach of this thesis is most apparent when looking at it as a whole. First, the research was conducted through close collaboration between academics and practitioners, guided by questions that concern real-world problems to which practitioners require timely guidance and practical solutions. Second, the research methods employed include a mixture of systematic literature mapping, systematic literature review, and qualitative case studies explored through interviews. However, it should be noted that much of the empirical data collection and analysis in this thesis was guided by interpretivism. This is because the pragmatist literature was found to be (oddly enough) less pragmatic in terms of offering guidance for less experienced researchers, since it may employ such a wide range of research methods. Furthermore, interpretivism enables a deep understanding of subjective experiences and social contexts and thus fits well with the aims of an empirical study within the social context of an organization (Wicks & Freeman, 2008).

In organizational theory, the interpretive paradigm suggests that there are shared multiple realities that rely on social constructs that are assigned subjective meanings through everyday language (Morgan, 1980). The end goal of the interpretive approach to research is not to depict an objective reality but to understand multiple subjective perspectives and their foundations. In organizational research, this translates into focusing on how organizational realities are created and sustained (ibid). Creswell (2014) also states that the goal of interpretive researchers is to rely as much as possible on the participants' views of the situation being studied. Typically, interpretive researchers choose an inductive approach, rely on qualitative methods, ask open-ended questions, and interpret what they find, acknowledging that their own experiences and background shape this interpretation. This latter point calls for reflexivity on the part of researchers, in which they acknowledge their own beliefs, judgments and practices during the research process and how these may have influenced the research. These matters will be discussed in the limitations section following the final discussion.

In the following sections, the research design and quality aspects of the appended papers will be outlined and discussed.

4.4 Research Design and Quality

This thesis explores the managerial challenges that large, established industrial firms encounter when implementing CE principles and how they address them, particularly through managerial controls. Additionally, it aims to provide insights into how management can facilitate radical circular innovation and support the circular transformation of incumbent firms. As the theoretical section highlights, this study poses research questions that have not previously been explored in depth, and thus it could best be described as exploratory. To achieve the aim of this thesis, the research design consists of a systematic literature mapping, a systematic literature review, and multiple case studies (see Table 3). Note that the case study organizations have been labeled *Tools*, *Forestry*, *Transportation*, *Mining*, and *Trucks*. These simple labels do not capture all the organizations' respective business activities and should be seen as a pragmatic way to refer to the different companies without revealing their identity. While *Transportation* may imply more services than *Trucks*, this is unintentional. They are all manufacturers that sell physical products, with varying degrees of service offerings. The more descriptive details about the case companies are found in the appended papers.

Table 3: Overview of research projects and connection to appended papers.

Paper	Aim	Cases	Method
A	Compiling and analyzing barriers to a CE transition by systematically mapping literature within the broader scope of barriers to circular economy, sharing economy, product-service systems, and collaborative consumption.	N/A	Systematic literature mapping (n=527)
B	Compiling and analyzing barriers to access-based consumption in a CE transition	N/A	Systematic literature review (n=22)
C	Describing the strategic implementation of CE at three incumbent firms and analyzing their experience in terms of exploration and exploitation	Tools Forestry Transportation	Semi-structured interviews (n=30), thematic analysis
D	Describing management control system interventions at three incumbent firms in light of CE and analyzing how they enable or impede CE transformations at those firms	Tools Mining Trucks	Semi-structured interviews (n=38), thematic analysis

Each method used will be briefly described in the following sections, focusing specifically on aspects that affect the quality of the research, as well as particular details that are not included in the appended papers, whereas additional reflections on limitations will follow the final discussion section.

Systematic literature mapping (Paper A)

A systematic mapping study compiles studies that address a subject of interest—often asking different research questions—and structures them into a system of categories (Okoli & Schabram, 2011). It is particularly useful to create this type of structure when dealing with complex or rapidly evolving fields of research where there are several different streams of discussion evolving in parallel. The resulting map is an inventory of papers—often in the form of a visual image or searchable resource—that gives an overview of a research area and can act as a baseline for subsequent systematic literature reviews. As the purpose is merely to give an overview, mapping studies use metadata: i.e., data that describe and provide information about other data (such as the study setting, design, interventions, and populations) and the paper in which the data appear (such as author, title, year, peer review journal, conference proceeding). For Paper A, the mapping was created by extracting metadata from abstracts of published studies. The work of James et al. (2016) and Petersen et al. (2015) was particularly helpful in structuring my method, which consisted of the following steps: create a protocol, select a search engine, decide on the search string, download the metadata (search results), screen abstracts for inclusion and exclusion, extract and classify metadata, and create a synthesis. The appended paper by Sopjani et al. (2020) outlines each step of the method in detail, justifies the decisions taken, and discusses some limitations of the research method.

Systematic literature review (Paper B)

A systematic literature review (SLR) should not be confused with the literature review found at the beginning of most empirical research papers. Instead, it is a stand-alone research project that uses existing studies to investigate a clearly scoped problem or question, following a predefined method. Following Fink (2014), an SLR must use a systematic methodological approach and explicitly explain the procedures followed, and it must be comprehensive in its scope so that it includes all relevant material and is hence reproducible by others who follow the same approach in reviewing the topic. Notably, the concern with reproducibility is heavily influenced by Fink's primary focus on SLRs of quantitative studies in the health sciences rather than qualitative studies in social sciences, which are impacted by human subjectivity (Okoli & Schabram, 2011). In contrast, Denyer and Tranfield's (2009) guidance for SLRs in management and organizational studies notes that they should be transparent, inclusive, explanatory, and heuristic. These aspects that affect research will be explained below in terms of how they relate to Paper B.

Transparency

Denyer and Tranfield (2009) argue that organizational and management SLRs, Unlike SLRs of medical studies, should not necessarily document replicable methods but instead should aid transparency. This principle was applied in Paper B primarily by providing a clear and detailed methods section outlining each step of our research process. Second, providing supplementary material containing detailed descriptions of the data sample and analysis supports traceability. Third, Paper B outlines its conceptual framework, which does not

eliminate researcher bias but does provide the reader with an understanding of our theoretical lens going into the data analysis.

Inclusivity

Inclusivity refers to the fact that organizational and management studies are interdisciplinary and heterogeneous by nature, and many researchers fail to disclose their processes for data collection and analysis. Denyer and Tranfield (2009) therefore oppose proxies for research quality such as journal rankings and argue that reviewers should use quality checklists and justify the inclusion and exclusion of particular studies. Paper B used the database developed in Paper A to select studies, in order to be as objective as possible. While both Paper A and Paper B disclose and discuss their inclusion and exclusion criteria, the selection of studies was significantly limited by the level of detail in abstracts and titles, as well as the selection of keywords, since our database built on our screening for these. Paper B did not use any proxy for research quality, such as journal ranking or number of citations. Instead, every study was judged on its own merits, taking note of aspects such as limitations, research approach, research method, and how the data was analyzed (see Figure 5).

AD	AE	AF
What is the aim of the study?	**How was empirical data collected?**	**How the data was analysed?**
The purpose of this paper is to examine the effects of inhibiting, motivating, and technological factors on users' intention to participate in the sharing economy, using Uber users as a case.	A self-reported online survey was conducted among Uber users in Hong Kong. A total of 295 valid responses were collected.	Using structural equation modelling using partial least squares (PLS) analysis
The authors argue that political competition is a critical ingredient that explains whether cities accommodate or ban ridesharing platforms and that this relationship is moderated in more populous cities and in cities with higher unemployment rates.	They collect the entire population of U.S. cities that Uber entered during the 2011–2015 period. From the data collection, they observed N = 208 U.S. local markets that Uber entered by the end of December 2015. The list of markets in which Uber operated as of December 2015 was drawn directly from the company's website. They augment this list with a slate of cities Uber had previously entered but exited by the end of December 2015. These cities that Uber exited were identified by searching all the company's official announcements on its website regarding an exit, complemented by extensive searching of all media coverage related to Uber for all U.S. cities over the 2011–2015 period in the Factiva, NewsBank, and LexisNexis databases.	To test our hypotheses, an ideal experiment would take two cities that are similar in all dimensions except political competition, have Uber enter each simultaneously, and then see how variation in political competition affects the regulatory outcome (ban or accommodate ridesharing platforms). Empirically, the autors used a logit model while controlling for many observables or by using a propensity score matching method while matching observables (e.g., city- specific or mayor-specific characteristics)

Figure 5: Example of how research quality was documented during the full-text reading for the SLR.

Explanatory

An SLR provides a synthesis that can be either aggregative or interpretative and explanatory. While the first method is a way of gathering descriptive reporting evidence that seeks to avoid or mitigate bias, the other two are creative methods in which the reviewer plays a more active role (Denyer & Tranfield, 2009). Therefore, an explanatory SLR should bring pieces from individual studies together to create new insights, with the aggregated results being greater than the sum of its parts. This approach was applied in Paper B, where the primary focus was on the themes developed from the analysis (i.e., the interpretive and explanatory analysis), supported by examples from individual studies. Again, the supplementary material provides for traceability and offers a descriptive analysis of the data.

Heuristic

Denyer and Tranfield (2009) argue that in management research, rather than presenting the truth or solutions in the form of valid evidence, it is more common that the outputs of systematic reviews are rules, suggestions, or pragmatic guides that may be useful in making progress towards a solution of a problem. In this vein, Paper B aggregated specific barriers to overarching themes and suggested relationships among these themes, as well as identifying areas that need further development for solutions. Therefore, this SLR describes the current state of knowledge on problems and points towards actors and actions that can be engaged to solve those issues.

Case studies (Paper C and Paper D)

Case studies can be used in many research areas but are particularly relevant when it comes to evaluative research that seeks to explain causal links, describe interventions, illustrate topics, and explore situations. In a case study, the researcher develops an in-depth analysis of a particular case, such as a program, event, activity, process, or one or more individuals (Creswell, 2014). In the words of Yin (2009), a case study is an empirical inquiry that investigates a contemporary phenomenon within its real-life context, especially when the boundaries between phenomenon and context are not evidently clear. This description pays special attention to the interest in contextual factors, which are not always captured by research strategies such as laboratory experiments, where control of contextual background variables is of pivotal importance. Furthermore, unlike in laboratory experiments, in a case study the researcher does not necessarily perform an intervention but instead studies real-life interventions. In the empirical studies included in this thesis, the main intervention of interest was the case companies' implementation of CE principles.

In a multiple case study, several cases are investigated within the same study. In other words, the same theoretical framework, research questions, and data collection methods are applied to different cases. Yin (2009) suggests a research strategy in which each case is treated at first as an individual case in isolation, and only in the final stage of the research is compared in a cross-case analysis with the other selected cases. This is the approach that Paper C uses. From an interpretive perspective, knowledge from case study research accumulates in a relative sense through the formation of ever-more-informed constructs and then describing and contrasting different constructs. The transfer of knowledge from one setting to another is done by providing indirect experiences, which is a characteristic of case study reports (Guba & Lincoln, 1994, p. 114). Case studies often use analytic generalizations, where a set of previously developed theories is used to contrast the empirical findings from the case study. This approach allows comparison among similar plausible rival theories to determine which are better supported and which are not (Yin, 2009). Theory, therefore, does not act passively as a background to the study but instead serves as an active vehicle to transform specific findings into something more generalizable. Insights and findings are not directly transferred from one case to another; instead, the comparison is mediated by theory. During the process of data collection, analysis, and paper writing, the research team always considered relevant theory in an iterative process, based on that the research topic is multidisciplinary and could be discussed from several perspectives. In practice, this was done through analyzing and collecting data concurrently, discussions at project meetings,

reworking the interview guides, and by experimenting with data coding, as will be discussed in greater detail in the following sections.

Qualitative interviews

Studying organizations requires studying the behavior of individuals or groups that interact with each other to perform activities intended to accomplish a common goal. This description understands organizations as a human social phenomenon, and thus it falls squarely within the field of the social sciences. Studying the behavior of individuals and groups offers many qualitative nuances, and data collection methods such as interviews, observations, questionnaires, and document analysis have become conventional when studying management and organizational theory (Eisenhardt, 1989). Researchers in this field frequently employ interviews (Gioia et al., 2013), as they encourage interviewees to articulate tacit knowledge which can then be transferred, evaluated, and hopefully improved on in the form of the aggregated knowledge the researcher accumulates. The goal of a qualitative research interview (from an interpretive viewpoint) is to see the research topic from the perspective of the interviewee and to understand why and how they have come to this particular perspective (Cassell & Symon, 2004).

Data collection

The primary form of data collection in this research was semi-structured, in-depth interviews, which were always guided by an interview guide that contained pre-determined questions building on extant theory and the research questions at hand. However, because they were semi-structured, the interviewer was able to modify how the questions were phrased and ask follow-up questions as required (Creswell, 2014). This data collection method is most suitable for exploratory research, given that the researcher learns more about the topic and the object of study as the research progresses.

Respondents were selected based on their different roles and responsibilities at each organization. Given that the researcher cannot determine these variables without profound prior knowledge of the organization, interviewees were selected in collaboration with contact people at each organization who had appropriate knowledge about the research study and about the organization's personnel: i.e., the interviewees were purposively selected (Saunders et al., 2009, p. 233). Because this is a non-probabilistic form of sampling, the number of interviewees at each organization did differ. However, from an interpretive viewpoint, it is not the number of informants that is crucial, since arriving at a universal truth is the aim of the research; rather, the study's aim is to compile individual accounts, viewpoints, and experiences relevant to the studied phenomena. All interviews during the thesis were conducted by the author, although sometimes the interview context included various constellations of one or two additional researchers. All interviews were audio recorded and transcribed verbatim.

Analysis of interview data

The interview data was analyzed using qualitative data analysis software, following a thematic analysis approach. Codes were created for both Paper C and Paper D, assigning meaningful labels that correlated with the studied phenomenon. The coding was grounded

in data and theory, and employed the same theoretical framework as the interview guide. However, the process used in Paper C was slightly different from that used in Paper D. In Paper C, the process was guided by DeCuir-Gunby et al. (2011) to develop a codebook, and Gioia et al. (2013) and Kitto et al. (2008) were helpful for the procedural rigor. Here, since there were multiple researchers involved in the data analysis, the initial focus was to create a codebook that would be validated by cross-checking and discussing each other's coding to arrive at a consensual interpretation. In Paper D, the author took on a more significant role in the data analysis, and the focus shifted from constructing a codebook that could be used by multiple researchers towards acquiring guidance from a structured process. Specifically, the four steps to thematic analysis outlined by Braun and Clarke (2006) were followed, and complemented by qualitative data analysis techniques suggested by Miles, Huberman, and Saldana (2014). These steps are described in detail in Paper D.

5. Summary of Appended Papers

In this section, the appended papers are briefly summarized by describing their aim, research method, main results, and contribution to the thesis. The author has had a significant role in all appended papers, although the distribution of work varied for each (see Table 4). While all attributed authors contributed to every paper, the portion of work has increasingly been led by the present author in order of A to D.

Table 4: Author contribution to the appended papers.

Paper	First Author	Co-authors	Description of work
A	Liridona Sopjani	Johan Arekrans, Rafael Laurenti, Sofia Ritzén	The planning, data collection, analysis, writing, and review of this paper were conducted jointly. All authors contributed equally.
B	Johan Arekrans	Liridona Sopjani, Rafael Laurenti, Sofia Ritzén	The planning, data collection, first analysis, and first draft were a joint product. J.A led the final analysis and subsequent revisions of the first draft.
C	Johan Arekrans	Sofia Ritzén, Rafael Laurenti	The planning of the project was done jointly, J.A conducted the data collection together with S.R. All authors were involved in the data analysis, but J.A was responsible for the majority and also led the paper writing and revisions.
D	Johan Arekrans	Sofia Ritzén, Susanne Nilsson	The planning of the project was done jointly, J.A led the data collection together with S.R., J.A conducted the majority of the data analysis and paper writing.

5.1 Paper A

Sopjani, L., Arekrans, J., Laurenti, R., & Ritzén, S. (2020). Unlocking the linear lock-in: Mapping research on barriers to transition. Sustainability, 12(3), 1034.

This paper describes the systematic development of a literature map concerning on barriers related to the transition to CE. The purpose was twofold: to classify published research on barriers, and to provide a searchable database for more rigorous systematic literature reviews in the future. This was achieved by reviewing the abstract from 527 publications and classifying each study according to a developed scheme. This classification scheme included the research paradigms (circular economy [CE], sharing economy [SE], collaborative consumption [CE], and product-service systems [PSS]), circular strategy (recycling, remanufacturing, component/product reuse, and access-based consumption), country, sector/industry/product, and research approach and method. The research questions below were used to guide the work:

- RQ A1: What are the publication trends with regard to barriers to the paradigms of CE, SE, CC, and PSS?
- RQ A2: Which are the dominant research paradigms and circular strategies in the literature to date in relation to barriers?
- RQ A3: What research approaches, methods, and sources of data have been used to identify barriers?
- RQ A4: How have barriers been conceptualized (the epistemological basis) in the literature?
- RQ A5: What is the contextual framing of barriers in relation to the level of analysis (e.g., industrial sector)?

The resulting research map is open source and also serves as an adaptive digital database for researchers to perform systematic reviews and contribute to the continued collaborative development of the map. Notably, other researchers have used this database for subsequent publications in scientific journals (see Försterling et al., 2023).

Some findings were particularly important in setting the direction for the thesis, specifically:

(1) Barriers related to management and organization at industrial firms seeking a CE transition are not well studied. In fact, we found surprisingly few studies focusing solely on barriers related to management and organization at industrial firms (7%) and only found in the paradigms of PSS and CE. However, it should be noted that this measurement is crude, since a significant portion of the dataset was also classified as either “business model design” (16%) or “more than one level of analysis” (24%)—categories that could encompass studies focusing on managing and organizing industrial firms.

(2) Recycling is the most studied R-strategy. In studies where a focus on a specific R-strategy could be identified (64%), the most frequently studied R-strategy was recycling. As recycling arguably does not require the same degree of radical change to business models or logics at

industrial manufacturers that other R-strategies do, this indicates a need to understand how to facilitate more radical circular innovation.

This paper contributes to the thesis by providing a broad overview of relevant research, identifying research gaps, building on extant knowledge, and positioning the findings of the thesis where relevant.

5.2 Paper B

Arekrans, J., Sopjani, L., Laurenti, R., Ritzén, S. (2022). Barriers to access-based consumption in the circular transition: A systematic review. Recourses, Conservation and Recycling

This paper is a systematic literature review aimed at compiling and analyzing barriers to a specific radical circular innovation for closing material loops, namely access-based consumption (ABC). A number of different literature streams have studied ABC, and Paper B synthesizes the streams of product-service systems (PSS), the circular economy (CE), the sharing economy (SE), and collaborative consumption (CC). The review builds on the systematic literature mapping of 527 articles in Paper A, selecting a subset of 50 articles based on a filtering of the mapping database. While several previous systematic literature reviews have discussed barriers to CE transitions, their scope has generally been limited: for example, to one specific paradigm (PSS, CC, SE, CE), product (such as clothing or mobile phones), or stakeholder (consumers or policymakers, for example). This review examines the common denominator of retained ownership: namely, the sale of *access* to a product rather than *ownership*, which represents a radical departure from linear production and consumption. The following research questions guided the work:

- RQ B1: What barriers to ABC have been empirically identified and reported in the literature streams PSS, CE, SE, CC?
- RQ B2: What theme of barriers are more prominent?
- RQ B3: What areas need further attention for the research and practitioner community to develop solutions towards overcoming barriers?

The barriers were examined through the perspectives of businesses, users, society, and multiple stakeholders. Through a full-text reading, a total of 289 barriers were extracted. Further analysis and categorization of the extracted barriers identified 17 themes. Because the primary focus of this thesis is business organizations, three themes related to barriers (BUS1-BUS3) from the perspective of business organizations are highlighted here:

BUS1. Business configuration, organizational structure, and culture inappropriate
BUS2. Partnerships, cooperation and engagement among stakeholders insufficient for ABC
BUS3. Product design, material, or technology inappropriate for ABC offerings

In summary, the analysis of all the themes and barriers found in the literature found that (1) the overall user experience of ABC and trust mechanisms between actors need to be better understood; (2) organizational aspects in traditional business need a system change; (3) regulation plays a fundamental role in making ABC work for business, society, and

sustainability; and (4) sharing risks and experimentation between actors for new learnings are necessary. The study recommends that future research gather experiences from ongoing experiments, identify best practices, and define viable mechanisms to facilitate circular behaviors for all stakeholders.

The three themes complement each other by distinguishing between internal organizing (BUS1), external engagement (BUS2), and technical issues (BUS3). While all the identified themes of barriers are relevant for a CE transition, a thesis cannot focus on all of them. Therefore, much of the following empirical work has related to "*BUS1 - Business configuration, organizational structure, and culture inappropriate,*" i.e. internal organizing. The paper contributes to the thesis by reviewing known challenges that incumbents may face when embarking on a radical shift in business logic from selling products to offering access, and it acts as a theoretical foundation going into the empirical studies.

5.3 Paper C

Arekrans, J., Ritzén, S., Laurenti, R., (2022). The Role of Radical Innovation in Circular Strategy Deployment. Business Strategy and the Environment

The aim of this empirical paper is twofold: first, to describe the actions taken by three large industrial incumbent firms following senior management decisions to adopt CE as a strategic issue, and second, to analyze how the deployment of circular strategies relates to organizing for exploration and exploitation in order to be competitive in the short term and deliver radical innovation in the long term. Two research questions guided the work:

- RQ C1: How are circular strategies deployed within large, established firms?
- RQ C2: How does the deployment of a circular strategy relate to exploitation and exploration in large, established firms and what are the implications for radical innovation?

A total of 30 semi-structured interviews were conducted. The interview data were coded in multiple rounds of thematic analysis using qualitative data analysis software. The study confirmed that all case companies had high ambitions for CE. However, it found the three firms employed diverse approaches to implementing and adopting CE, reflecting their different stages of maturity and the challenges they faced in the transition. When the activities and challenges the interviewees described are scrutinized from the perspective of exploration vs. exploitation, it was clear that incremental steps towards CE were possible by exploiting firms' existing resources and capabilities. In contrast, when the firms described exploration activities, they all pointed to significant challenges in achieving radical circular innovation. The analysis found that these challenges primarily related to:

(1) approaches to normative management,
(2) how the innovation ecosystem was engaged,
(3) how goals and metrics related to CE, and
(4) resources and coordination regarding the CE initiative.

Furthermore, deploying a circular strategy challenged several established norms at the organizations. Based on the analysis, normative management issues for circular strategy implementation also bear a strong relationship to achieving radical innovation (e.g. fostering experimentation, risk-taking, and questioning business-as-usual). In addition, it was found that many of the achievements in circular innovation to date had been driven by passionate individuals operating outside of their usual roles and responsibilities, who are left to manage the colliding interests between the linear business and their circular projects.

One of the key findings concerns goalsetting and measurements. Specifically, the interviews uncovered many opposing views and experiences with the use of goals, measurements, and managerial controls as a means to drive change at firms. The paper suggests that goals and metrics need to be considered in the deployment of circular strategies because they give more specific direction for the strategy, and they are fundamental control mechanisms in organizations. Two of the cases had significant incompatibilities between the CE strategies implemented and their existing goals and measurements. This, in turn, led to significant tensions between business as usual and driving the CE strategy, which risked favoring the status quo due to the pursuit of incompatible goals. The cases also revealed opposing views and different approaches concerning how goals and measurements should be used when implementing CE strategies, with possible implications for radical innovation. One view considered goals to be valuable for alignment, learning, incentives, and spurring creativity for experimenting with CE principles, while the other view primarily pointed towards the risks of sub-optimization and challenges in quantifying R-strategies. Consequently, the paper identified a need for further research to understand how goals for circular innovation can drive organizational change, how they impact ideation concerning CE, and their compatibility or incompatibility with existing governance systems and measurements.

This paper adds to the thesis by gathering empirical evidence on the challenges faced by incumbent firms in CE transitions and by highlighting key issues for achieving radical circular innovation. The findings in this paper were essential in developing the remainder of the thesis, as they indicated that MCSs could be both drivers of and barriers to CE adoption, a possibility that was investigated further in the following paper.

5.4 Paper D

Arekrans, J., Ritzén, S., & Nilsson, S. (2023) Tensions in Management Controls; Enabling radical innovation for a Circular Economy. [Submitted manuscript]

This empirical paper aims to provide greater insight into how firms utilize managerial controls to strengthen CE principles within their organizations. It also highlights critical tensions in the management control system following the two contrasting logics of linearity and circularity. To guide the work, two research questions were formulated:

- RQ D1: What MCS interventions do incumbent firms make when implementing CE?
- RQ D2: How can MCS enable or impede the CE transformation of an organization?

The data collection consisted of 38 semi-structured interviews at three large industrial incumbent firms, which were transcribed and coded using thematic analysis. In summary, we show how CE implementation is still nascent at the investigated companies. However, while it is not yet fully integrated into all aspects of the MCS, we identified how the MCS still influences the CE transformation. The most prevalent enabling factors in the studied companies concern managerial controls that favor (1) integrated work with sustainability, (2) the integration of different organizational functions, (3) the legitimacy of CE, (4) experimentation, (5) a broad scope of organizational responsibility, and (6) a long time horizon, to facilitate radical circular innovation. In contrast, the analysis also indicates that the MCS can lead to a linear lock-in depending on the following factors: (1) incremental innovation, (2) optimizing for efficiency, and (3) marginalization of CE. In this case, action controls and results controls strongly influence the most dominant lock-in factor of incremental innovation compared to cultural controls.

The paper adds to all three research questions that the thesis poses by adding rich empirical evidence on the challenges faced at large, established industrial manufacturers (RQ1), explicitly investigating radical circular innovation (RQ2) and how management control systems can act as both barriers and enablers of CE transformations (RQ3), as will be discussed in greater detail in the following section.

6. Analysis and Discussion of Paper Contributions

Before discussing the contributions of the different appended papers in the following section, a brief overview of the research questions and contributions of each paper is presented in Table 5 below.

Table 5: Research questions and contribution of papers.

	RQ1	RQ2	RQ3
	What challenges does radical circular innovation pose for large, established industrial firms?	What aspects of managing innovation are critical for achieving radical circular innovation at large, established industrial firms?	How do management controls facilitate or impede circular economy transformation at large, established industrial firms?
Paper A Literature mapping	Literature mapping of barriers to CE transitions to establish a broad overview of prior research on these challenges.	-	-
Paper B Literature review	Literature review to analyze barriers to ABC, which represents a radical circular innovation for incumbent firms.	-	-
Paper C Empirical	Case studies of challenges of CE implementation and strategy deployment at three incumbent firms	Analysis of how exploration and exploration were managed with regard to radical circular innovation	Identification of tensions between CE adoption and existing normative management practices
Paper D Empirical	Case studies of managerial challenges related to CE adoption and MCS at three incumbent firms	Analysis of how MCSs facilitated or impeded radical circular innovation, and theorization on the implications for the future development of MCSs	Analysis of tensions and conflicts between existing MCSs and CE at three incumbent firms and how an MCS to enable CE adoption might be designed

The research questions and appended papers represent a journey of studying an evolving phenomenon that is complex to tackle for both researchers and industrial practitioners. The doctoral studies started from a relatively broad scope, trying to understand what challenges a CE transition posed for incumbent firms. As more insights were gathered from existing research and empirical data, it became increasingly relevant to ask narrower questions (i.e. the use of managerial controls). This "funneling" process is illustrated in Figure 6.

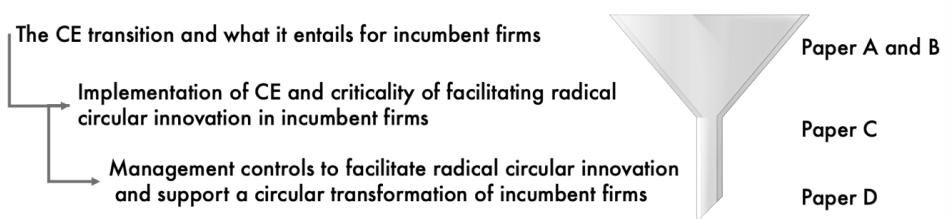


Figure 6: Illustration of the scope and funneling of the appended papers.

The primary contribution of the thesis is anchored in its answers to RQ2 and RQ3, with RQ1 serving as a foundational precursor. Consequently, the following analysis and discussion section is focused on RQ2 and RQ3. Notably, as Table 5 indicates, identifying challenges is an underlying theme of all the research papers, as each appended paper adds its own perspective to a more holistic understanding. As Paper D explicitly focuses on managerial controls and Paper C on exploration and exploitation, the ambition of the following section is to both revisit the findings from these papers individually and to shift the theoretical lens between the two in a form of meta-analysis. For example, while the research design of Paper C did not focus on managerial controls, the analysis in this paper nevertheless revealed relevant findings on this topic that are revisited through this lens of managerial controls. The concept of cognitive frames is used to give structure to this analysis and bridge between the two studies, as will be explained in the following section.

6.1 Viewing the Circular Economy Through a Radical or Incremental Frame?

This thesis starts with a simple idea: we must make significant changes in how we make and use "stuff." In this vein, CE has gained much attention from experts, practitioners, and policymakers as a feasible way forward in which economic growth does not have to deplete natural resources. However, here is the catch: we do not really know how large firms are organizing to put CE into practice and whether they are actually pushing for major innovations to drive CE transitions or are instead making smaller peripheral tweaks to comply with regulation; if the latter is the case, then why? We need to understand how and why decisions relating to CE are made at the managerial level and how these decisions influence radical circular innovation at large organizations. Therefore, to discuss the rationale behind individual decisions, the idea of cognitive frames is applied. Then, attention is brought to the interaction between managerial controls and cognitive frames, and how specific frames gain traction within an organization.

Radical and incremental framing of the circular economy

Throughout the interviews conducted for this thesis, interviewees at the different case firms described situations in which groups of people targeted the “low-hanging fruit” of R-strategies in ways that only departed slightly from previous practices. Other interviewees emphasized R-strategies that were more radical to the firm. This is one source of tension observed throughout the research process: i.e., whether CE adoption requires radical innovation or incremental change, or perhaps both. For example, when informants in Forestry were asked to describe how the company relates to CE, their responses could be classified into two different perspectives. The first group emphasized the role of Forestry in a circular *bioeconomy,* emphasizing how the company provides renewable materials for its customers and thereby fulfills an important role in achieving circularity merely from the type of materials it produces and uses. That is, CE could be achieved by Forestry’s exploitation activities. The second group agreed with this viewpoint but also highlighted the importance of new business models. They pointed out the missed potential in employing additional R-strategies, such as reuse, and the consequent changes in product design that these strategies

would necessitate. From this perspective, exploration would be essential for CE implementation. It is essential to highlight these conflicting viewpoints, which were a recurring pattern in the studied organizations, since exploration and exploitation have different requirements.

In complex decision-making situations, people have limited capacity to process all available information: i.e., our ability to make decisions based on rational analysis is bounded. Building on this to discuss how individual managers interpret situations, the concept of cognitive framing is utilized. Figure 7 revisits an illustration from Walsh (1995) and applies it to the context of this thesis. This figure illustrates the interaction between the elements involved in managers' framing of CE transitions as either radical or incremental; this perspective is used to structure the following analysis.

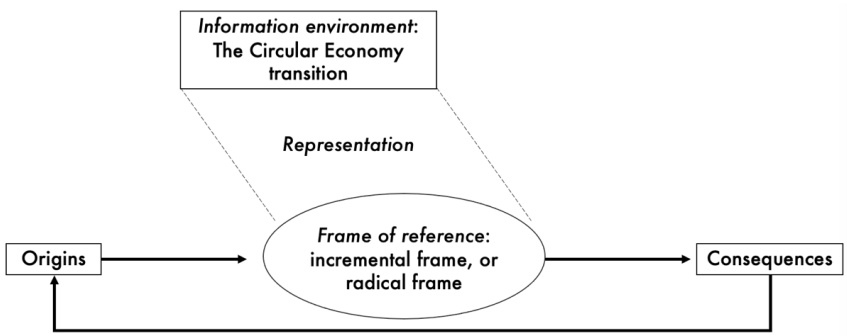


Figure 7: Framing the circular economy transition through radical or incremental cognitive frames, based on Walsh (1995).

The *information environment* where the studied firms find themselves is the transition towards a circular economy. For firms to reduce the complexity of the CE transition, the phenomenon is processed and interpreted through frames of reference, which can act at the individual, group, or organizational levels. In the empirical studies, there are several indications of how managers were framing CE. This thesis suggests categorizing these framings as either radical or incremental. The *frame of reference* in Walsh's (1995) model, is thus incremental or radical. Viewing these framings dichotomously as either incremental or radical is a simplification, but this view is grounded in the differing views of the interviewees. Here, it is worth noting that in the appended papers there is no clear-cut categorization of the cases as following either radical or incremental framings. Instead, one should remember that framing can operate at various levels, including the individual, the team, the business unit, and the organization. While the appended papers refer to the organizational level when referring to cases from a macro perspective, this mirrors the collective dominant framing (Kaplan, 2008) following the sensemaking process done by individuals. The interviewed informants often reflect upon several these different levels during the interview. The same is true for the *origin* of the different framings, which can relate to the firm as a whole, its subunits, or individuals. To exemplify the origins, in Paper C, Forestry's paper division found itself dealing with a particularly challenging scenario as the need for newsprint declined dramatically. In this context, division managers looked to

CE as a way to find radical new business opportunities, a perspective that contrasted with several other informants from other divisions. At Tools, professional customers had expressed significantly less interest in circular offerings than consumers, which led different divisions to frame CE differently. Ultimately, these differences in framing are a source of tension between individuals, teams, and divisions whose implications of which will be further explored and illustrated. These frames thus have different *origins* and ultimately lead to different *consequences* in the organization's behavior concerning CE, as will be further discussed below. To outline how this general idea of cognitive frames is applied in the context of this analysis, see Table 6.

Table 6: Description of the elements relevant to cognitive framing applied in a CE context.

	General description	In the context of CE
Information environment	The phenomena that the subject has interpreted or related to	The circular economy transition
Frame of reference	The cognitive frame or knowledge structure through which the subject processes the phenomena	CE requires either (1) radical or (2) incremental changes
Consequences	Decisions made and actions taken based on the processed information	E.g. how is CE is operationalized, the firm's ambitions with respect to CE, and how the organizational change is facilitated
Origins	The foundations on which the frame of reference is built and the reasons behind it	E.g. what the present business environment looks like and what the drivers for CE adoption are

Incremental framing: Viewing the circular economy as an incremental improvement
Firms can introduce an immense number of incremental improvements to existing products and services to improve circularity, meaning not departing significantly from current practices (Dewar & Dutton, 1986). For example, firms can slow down resource flows by making minor changes in component design so their products are more repairable or by adopting more durable materials to prolong their product lifespans. They can also make changes in auxiliary materials, such as those consumed during manufacturing or employed in product packaging. Such "end-of-pipe" technical changes or eco-efficient optimizations to current practices can help firms reduce waste creation while avoiding unnecessary costs (Kennedy et al., 2017). Individuals who embrace this view can be said to hold an incremental framing of CE. Typically, these sorts of innovations can be achieved using the firm's existing technology and capabilities and do not put new demands on the business model. This type of reasoning does not fundamentally question the paradigm of linear production and consumption or the organization's role in society.

Radical framing: Viewing the circular economy as a radical shift
At its essence, CE proposes ways to decouple economic growth from resource utilization. This implies an industrial system that is not built on maximizing the throughput of manufactured material stock but instead on maintaining the quality and quantity of that material stock (Stahel, 2019). Therefore, product manufacturers must take greater

responsibility for what happens with their products beyond the point of sale, ensuring that they maintain their value over time. This contrasts strongly with the traditional linear economy, in which the consumption of goods is a fundamental cornerstone of the financial viability of business models. Individuals adopting a radical framing recognize this discrepancy at the systemic level, as well as its implications at the local level. Changes made to products and business models are seldom justifiable in terms of circularity or economic feasibility if they are not considered in an integrated fashion. For example, a remanufacturing operation's economic feasibility depends on decisions made early on in the product design process. Referring back to the opening story of the 120-year-old lightbulb as an example, a pay-per-lumen service business model coupled with a product designed for longevity could have been an interesting alternative to planned obsolescence.

Table 7 outlines the origins and consequences of radical vs. incremental framings, which will be discussed in detail in the following sections.

Table 7: The origins and consequences of incremental vs. radical framing of CE.

		Incremental framing *Minor improvements to business as usual*	**Radical framing** *Significant changes to business as usual*
Origins	Characteristics of the firm, its value proposition, and the industry as a whole	Success in business-as-usual Stable business environment Cost-leadership strategy	Turbulent business environment Technological shifts Differentiation strategy
	Main drivers for CE adoption	Legislative demands Stakeholder expectations Short-term performance	Sense of urgency Business potential Long-term relevance Leadership in sustainable innovation
	Underlying business logic	Current business stable Minimal efforts to remain relevant	Current business is challenged Radical measures are required to change
Consequences	Ambition with CE	Reduce environmental footprint Improve economic performance Greater resource efficiency	Finding new business opportunities to replace business-as-usual
	Operationalizing CE	Targeting R-strategies that fit with existing business and sustainability practices Narrowing resource flows Recycling	Targeting various R-strategies Narrowing, slowing, and closing resource flows
	Means to reach CE ambition	Monitoring Controlling Predicting Reducing Preventing	Knowledge creation Exploring Ideating Experimenting Collaborating

Origins of radical and incremental framing

A CE transition does not require the same changes for all organizations. Instead, the extent and nature of changes are influenced by factors intrinsic to each organization and its external business environment. It is here argued that these organizational-level factors contribute to the origins of the framings made by individuals and teams. These factors include the firm's current value proposition, its capabilities and business environment, and the legacy systems and practices that have shaped its business-as-usual operations over time. Origins of the two contrasting framings will be outlined in the following section, anchored in data from the empirical studies (see Table 7 for an overview).

Business environments characterized by stability and continuity experience particularly robust inertia from established business practices. The self-reinforcing nature of organizational learning makes it more attractive for successful organizations and individuals to keep their current focus (Leonard-Barton, 1992; Levinthal & March, 1993). In such settings, the rationale for adopting CE principles may stem from external pressures and expectations, including stakeholders and public opinion, as well as anticipation of impending legislative mandates (i.e. getting "license to operate"). These organizations are driven by concrete demands for change that translate into a problem-oriented type of innovation (Cyert & March, 1992). Typically, this results in a focus on short-term performance improvements and an incremental approach to integrating CE practices into their operations. The aim is to align with expectations and ensure ongoing business viability without compromising or cannibalizing their current business activities.

In contrast, in dynamic or turbulent business environments, organizations may be more inclined to actively seek out and capitalize on radical new business opportunities. The turbulence in such environments may stem from factors that are not directly related to a CE transition; sources of such turbulence include rapid technological advancements, shifts in consumer behavior, and heightened competition within the industry. In these contexts, firms view CE not merely as a response to external pressures but rather as a strategic avenue for identifying and exploiting business opportunities. These organizations see CE as a means to secure their long-term relevance and resilience, embracing the concept not as a necessary adaptation or peripheral add-on but as an intrinsic part of their differentiation strategy.

These different origins are reflected in the studied cases. The Transportation, Tools, and Forestry firms all acknowledged CE as a strategic issue and had begun implementing changes in their organization in response. Transportation and Tools can arguably be classified as holding "visionary" sustainability strategies (Baumgartner & Ebner, 2010), as interviewees described the profound changes they were making to integrate sustainability within all business activities. In addition, interviewees and communications from senior management acknowledged that the business-as-usual scenario was not sustainable from an ecological point of view. Transportation found itself in a turbulent industry that was experiencing major disruptions in terms of technology and actors in its business environment. This firm acknowledged the impact its existing products and operations had on environmental sustainability. From a strategic perspective, these factors combined to

create a clear sense of the need for a radical shift. In this shift, significant business opportunities could be reaped by radical circular innovation, often relying on technology shifts in combination with an increased focus on servitization. In comparison, most of the departments at Forestry and Tools were experiencing comparatively less turbulent business environments. However, much of Tools' product portfolio was undergoing electrification, a radical technology shift in an increasingly competitive market segment, rendering the firm's previous strategic advantages and core capabilities less valuable. Both Tools and Transportation had a long history of design-for-manufacturing and the production of high-performance internal combustion engines—competencies that may conflict with the requirements of CE and could become obsolete following the electrification of their product portfolios. Their traditional competencies thus weighed against certain kinds of innovation (Hall & Vredenburg, 2003). This contrasts with the situation at Forestry, where all interviewees viewed a CE transition as favorable for their business because their value proposition was already based on renewable materials and thus CE was a clear "fit" for the firm.

The firms Trucks and Mining were also undergoing a significant shift in terms of electrification, and there, CE was viewed as holding business potential. This was coupled with the general public's negative views of the environmental sustainability of the mining and trucking industries, leading stakeholders to pressure these firms to find viable options to remain relevant (i.e. license-to-operate). Trucks and Transportation both also anticipated forthcoming legal requirements that would require them to take greater responsibility for the heavy trucks they put on the market, increasing the attractiveness of service-based business models such as vehicle-as-a-service or transportation-as-a-service, with the firm retaining ownership of the products.

In sum, the case studies illustrate how organizations in different industries and markets were undergoing profound shifts in their business models, technologies, and strategies in response to external pressures, strategic needs, and sustainability considerations. The firms exhibited different levels of alignment or conflict between CE principles and their current value propositions and competencies. These organizational-level characteristics and drivers are some of the elements that influenced whether individuals and teams framed CE implementation in terms of radical innovation or incremental change. While the factors outlined here are at the organizational level and relate to the interface between the organization and its business environment, later sections will explore how management controls are also an origin for different framings.

Consequences, tensions, and conflicts associated with radical or incremental framings

Whether individual managers frame CE as radical or incremental carries significant consequences when considered in an organizational context. Specifically, building on the idea of sensemaking (Weick et al., 2005) and dominant framings (Kaplan, 2008), it is here argued that an organization's ambition, operationalization, and means to achieve CE are the results of its dominant framing. This is indicated in the overview presented in Table 7. In other words, following the sensemaking process, one particular dominant framing (radical

or incremental) will come to prevail as a guide for individual actors. However, when a dominant framing has not yet emerged, it can lead to inertia and the deferral of decisions. In addition, the contrasts between the framings can lead to tensions between different actors if these framings continue to coexist. Lastly, decisions made based on different framings can lead to conflicts in operational activities if two decisions are not compatible. In the context of CE, this sensemaking process is further complicated by the multifaceted nature of CE and the fact that it has been defined and conceptualized in several ways (Kirchherr et al., 2017). Consequences of the two contrasting framings will be outlined in the following and then anchored in the internal organizational tensions and conflicts found during the empirical studies.

One consequence of adopting an incremental framing of CE is a focus on aligning CE principles with sustainability practices and goals already in place as part of the effort to reduce the environmental footprint of a company's operations (see Table 7). Typically, these efforts revolve around improvements to resource efficiency and recycling: Resource efficiency involves using fewer materials per product, thereby minimizing waste and resource consumption. Recycling involves incorporating recycled materials into products or designing products to be more easily recyclable. Both strategies have been criticized for not fundamentally altering the rate at which resources flow through the economy (Bocken et al., 2014; Braungart, 1998; Stahel, 2019). In line with the logic of efficiency, measures associated with an incremental CE framing often include reduction (e.g., reducing resource consumption), prevention (e.g., preventing waste and pollution), and monitoring and control (e.g., keeping track of environmental metrics). These actions are oriented toward creating tangible (and often measurable) improvements in the sustainability of the existing portfolio of products and services and primarily aim to reduce negative environmental impacts without undertaking a complete overhaul of the business model.

In contrast, a radical framing of CE treats the concept as novel and distinct from existing sustainability practices. Here, CE is not seen as an extension or alteration of established practices but rather as an innovative approach to business, as we see in Table 7. Therefore, one consequence of CE in a radical framing is an emphasis on exploration activities, since the organization needs to venture into uncharted territory in the search for new circular business opportunities. This triggers a need for knowledge creation and training, as employees must acquire new skills and understanding of the CE logic in order to ideate a diverse set of R-strategies. Because this sensemaking process can be slow and may entail a difficult process of “unlearning,” the radical framing many times leads firms to seek collaborations and partnerships outside their organizational boundaries, seeking external expertise and resources to facilitate radical circular innovation.

At all the case firms, informants shared examples of existing projects and opportunities where current resources and capabilities were considered sufficient for achieving some improvements in circularity:—i.e. exploitation—an approach in line with an incremental framing. While there most certainly are several incremental improvements that can serve as important steps in a more comprehensive CE transition, it is important that these

incremental changes do not overshadow other ambitions. In this vein, informants also shared examples of tensions over whether existing achievements in areas such as recyclability, improved manufacturing processes, or slight changes to packaging could be viewed as already contributing to CE, and worryingly, create reluctance for exploring endeavors with a higher circularity potential. This meant that their CE ambition was linked to corporate sustainability activities that were already underway and had a good fit with a narrow selection of R-strategies and therefore did not require exploration. Seeking to improve on activities that were already somewhat familiar to the firm meant that *doing better* needed to be proven in the form of activities that could be monitored and controlled as a way to offer feedback in the short term. Therefore, this relates to the problem that Levinthal and March (1993) raise, where fast-learning and successful firms are overly responsive to the early feedback that exploitation activities generate, risking a premature shift of resources away from exploration activities that are associated with uncertainty and risk.

Decisions made based on different framings can lead to conflicts in operational activities, as found in the case studies. Clearly, as has also been seen in prior research, radical circular innovation made demands on both business model and product design (den Hollander et al., 2017; Rashid et al., 2013). Except for Forestry (likely due to their different range of products and services compared to the other cases), informants in all cases described a historical focus on design for manufacturing (favoring speed of assembly and mass-manufacturing) and, in some cases, design for recyclability as well. Here, interviewees described how various design choices that favored mass-manufacturing or even recyclability hampered other R-strategies such as repair or remanufacturing, even though these strategies offered a higher circularity potential. Therefore, a shift towards circular business models would lead to product design requirements that contrast starkly with those set by the linear economy, in line with what den Hollander et al. (2017) outline. At Transportation, one informant offered the example of new design requirements for an automotive alternator that were introduced to improve the longevity of a component incrementally. As the informant explained, the design choices based on these new requirements were detrimental to the component's remanufacturing or repair, as they included welding; this conflicted with the company's ambitions for remanufacturing, which was managed in another department, pointing to local rationality of sub-unit goals (Cyert & March, 1992). Altogether, the requirements for knowledge creation and changes in product design and business models needed to achieve radical circular innovation are not easily justifiable under an incremental framing, and decisions made through different framings may lead to conflicts.

The CE literature emphasizes the need for partnerships, cooperation, and engagement of stakeholders. While the case studies in this thesis primarily investigated challenges *within* organizational boundaries, several informants in all cases emphasized that new strategic partnerships and closer collaborations with existing partners and suppliers would be essential to achieving their firm's ambitions in CE. At Forestry, circular innovation was described as often involving collaborations with other firms further down the value chain than what the company was accustomed to doing, or with new actors that were "born

circular" (Briguglio et al., 2021) who saw opportunities for closing material loops. However, comments from interviews at all case firms highlighted significant challenges with these types of intensified or new collaborations in exploration activities for radical circular innovation. Forestry described dependencies on local infrastructure, actors, and regulations that required coordination and alignment within and beyond organizational borders in order to facilitate circular business. Tools and Transportation described "old habits" that favored closed, in-house product development processes. Mining pointed to their low production volumes, which meant they had little influence over suppliers when discussing new requirements. Trucks stressed that even though they were making efforts to include traditional suppliers in the development process and engage them in developing solutions rather than components, many suppliers were said to be reluctant to change.

In brief, at the case companies, exploration activities for more radical circular innovation met with significant challenges and were outnumbered by exploitation activities. This finding fueled a need to investigate further the origin of these framings, as well as the mechanisms that managers could use to facilitate radical circular innovation. A bundle of various mechanisms at play shapes decisions and actions in organizations. Specifically, management control systems (MCS)—comprising action, result, and cultural controls—influence the behavior of employees and can be modified and adapted by management within an organization (Merchant & Van der Stede, 2017). Paper D examines how firms utilize MCS when implementing CE and the conflicts that emerge with their legacy MCS. Building on the challenges found in Paper C and investigating additional challenges in Paper D, particular attention is paid to how radical circular innovation can be facilitated or hampered by MCS interventions and legacy, as will be discussed in the following section.

6.2 The Management Control System and its Relation to Framing

To recap, building upon the idea of sensemaking (Weick et al., 2005) and cognitive frames (Walsh, 1995), it is suggested that when organizations implement CE, managerial decision-making tends to follow the logic of one of two framings: either a radical one that calls for significant changes or an incremental one focused on gradual improvements. Now, the analysis of the cases shows that each framing has its own set of implications for how CE principles are operationalized and integrated by individuals and across divisions in an organization, as well as for the types of managerial controls deployed (see Figure 8). This thesis adds the perspective of the management control system as a mechanism to influence the dominant framing (Kaplan, 2008) of CE and to overcome organizational barriers faced in CE adoption. In particular, studying how the management control system can support or impede the implementation of radical circular innovation at large industrial incumbent firms.

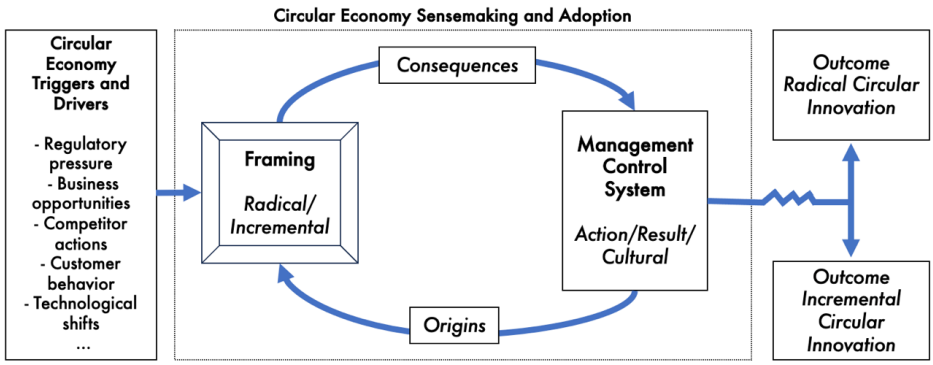


Figure 8: Relationship between CE drivers, cognitive framing, management control system, and outcomes.

As the figure shows, MCS may not only be an instrument for implementing and adopting CE but can also be a contributing factor (i.e. an *origin*) in how CE is framed. This relationship between MCSs and framing is embedded in organizational learning (Cohen & Levinthal, 1990; Levitt & March, 1988). Specifically, it is argued that the MCS is an artifact or manifestation of past experiences and actions that guide decision-making. This relationship is also self-reinforcing, which means that when existing managerial controls fit with a particular CE framing, it creates a positive feedback loop, and when information does not have a good fit with those controls, it is discarded or reduced.

In this model, framing and the MCS are interlinked and work together to influence the understanding and adoption of CE principles in organizational contexts. Understanding these interdependencies is proposed to be crucial for fostering coherent and effective implementations of CE principles and for navigating the inherent complexities and nuances of organizational learning and decision-making processes in order to achieve circular innovation. In particular, it is suggested that since the MCS can be deliberately designed and changed through interventions, it stands as a crucial mechanism for the successful adoption of CE in organizations. Furthermore, it suggests that an organization's MCS plays an essential role in influencing which framing of CE becomes dominant, thus potentially alleviating the tensions and conflicts that result if different individuals or teams within the same organization understand CE using contrasting framings.

In the following sections, the empirically studied management control origins, interventions, and consequences will be discussed, categorized after action controls, result controls, and cultural controls (Merchant & Van der Stede, 2017).

Action controls

The function of action controls is to provide guidelines for what should be done and to create boundaries for what should not be done. This can be achieved through process definitions, organizational structures, job descriptions, and resource allocation, to name a few (Merchant & Van der Stede, 2017).

Examining the findings from the case studies reveals instances of new roles and responsibilities that emerged after companies launched their CE initiatives. For example, new positions were created, such as the Director of Circular Economy at Forestry, the Circular Target Lead at Tools, and the establishment of the distinct organizational function of Circular Operations and Solutions at Trucks. These roles reflect deliberate efforts to streamline and coordinate CE initiatives across different divisions within each organization. Although the implementation processes were still in the relatively early stages, which makes it difficult to accurately assess firm-level performance, these new roles and responsibilities yielded several success stories at the project level and helped alleviate some of the challenges that interviewees reported experiencing.

At Forestry, introducing the Director of Circular Economy helped the organization effectively allocate resources to smaller divisions that were struggling with CE-related projects. Similarly, at Tools, a small group-level team was pivotal in guiding the initial sensemaking process and crafting a governance structure for their CE implementation. This entailed collaboration in formulating joint ambitions; and because the team incorporated representatives from each division, these ambitions were disseminated and well-anchored when they were subsequently operationalized at the respective divisions. This approach allowed the firm to target a broad portfolio of R-strategies. The case of Transportation provides a negative counterexample; at the time of the interviews, several members emphasized the lack of a joint CE framework, a designated organizational role, and resources dedicated to coordinating CE implementation. Instead, the bulk of CE coordination relied on the initiative of a single individual acting outside their regular role, which may not have been viable over time and was not a strategy that could effectively drive large-scale changes. This suggests that such newly established roles, responsibilities, and organizational functions were indispensable in facilitating the sensemaking, coordination, and allocation of resources essential for early-stage CE implementation and adoption.

Fundamental questions concerning the scope of organizational activities also fall within the category of action controls. For instance, Tools identified that novel approaches to reaching customers were a prerequisite for CE, highlighting limitations of their current business model, in which retailers were the primary location for transactional sales between producers and consumers. This setup was considered inadequate for achieving radical innovation, as it placed saddled retailers with the burden of customer onboarding and relaying customer insights back to producers. These types of concerns call the scope of organizational activities into question: i.e., should the company have more direct customer relationships and channels for reaching the end user? Similar considerations arose at Transportation and Trucks, where some felt that radical circular innovation would necessitate assuming responsibility for a larger portion of the value chain. Effectively, this could also mean taking on the business of their current customers. Tools experienced challenges in their pursuit of novel partnerships and collaboration in developing circular business models. The firm encountered negative experiences associated with internal legal processes and heavy documentation burdens related to intellectual property rights tailored

to traditional larger suppliers and partners. These challenges delayed riskier collaboration with the smaller "born circular" startups that radical circular innovation relied on.

In summary, action controls encompass a broad spectrum of controls, including roles, responsibilities, organizational structures, and operational processes within these companies. The analysis of the case firms made it evident that existing action controls tended to favor incremental innovation over radical innovation, since they often serve to mitigate risk and minimize variability. Nevertheless, interventions in action controls also enabled the companies to achieve greater integration of sustainability into their operations. Moreover, they brought forth new organizational functions and roles that directly facilitated the alignment and coordination of CE efforts throughout the organization, creating the essential conditions for the adoption of CE. Action controls can thus be designed to support both incremental and radical framings.

Results controls

The use of results controls at large and well-established firms, particularly with decentralized structures, is a well-documented practice for achieving alignment among different organizational units (Merchant & Van der Stede, 2017). Nonetheless, using results controls is particularly intriguing for firms that are seeking to adopt CE, for several reasons.

First, the challenge of quantifying CE performance has been a focal point of research over the past decade (Saidani et al., 2019), underscoring the inherent complexity of circular business models and R-strategies. Second, as found in Paper B, there is convincing evidence suggesting that results controls that follow a linear logic might hinder the successful implementation of CE initiatives. Third, results-driven behavior may lead to the pursuit of efficiency over innovation if result controls are not properly considered (Merchant & Van der Stede, 2017). Therefore, a critical investigation of the use of results controls to foster radical circular innovation becomes particularly important.

Results controls such as goals and measurements play a pivotal role in operationalizing a firm's strategy effectively. Interestingly, in the studied cases, various perspectives on the use of goals and measurements within the context of CE strategy deployment emerged (some of them conflicting). Some informants argued that the absence of clear goals and performance indicators created insufficient incentives for the organization to adopt CE. However, others believed that not setting goals allowed time for exploration activities in the initial stages of implementation. Some informants, notably those at Transportation and Forestry, avoided setting specific CE goals and performance indicators for fear that efforts would be misdirected if controls were improperly used or if good performance was incorrectly defined. This reasoning aligns with McGrath (2001) and Hansen et al. (2018), who suggest that early goals and exploitative routines may truncate the learning process needed to achieve the goals. However, within the same organizations, the opposite viewpoint was also voiced. Informants believed a lack of clear goals would hinder coordination—an essential enabler for radical circular innovation requiring extensive interdepartmental collaboration. Further, some interviewees felt that the lack of goals meant that senior management viewed CE as

less important than other ambitions. Svensson and Funck (2019) have also highlighted how CE objectives serve as internal communication tools that align the organization and signal top management's priorities.

Adding to the complexity of results controls, the empirical cases indicated a reliance on existing goals and metrics rooted in linear logic, both at the strategic and operational levels. Respondents highlighted traditional business metrics such as short-term profit and sales volume, which tended to promote short-term thinking and impede investments in long-term sustainability objectives. Specific performance metrics were found to be directly at odds with CE. For example, at Transportation, the share of new vehicle registrations was closely monitored as an indicator of market success. This control favored the production of new vehicles, and informants said it took precedence over other measurements such as aftersales maintenance services or remanufactured components when it came to decision-making and reward structures. As such, it was not only the performance metric per se, but management did pay more attention to this control metric. Similar examples were found at Trucks, Tools, and Mining, where metrics such as productivity, production, and profitability were prioritized by top management and closely monitored. This posed a challenge for CE, as radical circular innovation may not initially be profitable, and efforts to dematerialize, for example, went against the traditional production and consumption logic.

Nevertheless, some informants viewed results controls positively as drivers for CE implementation based on their experiences of having implemented them. For instance, Tools set a corporate goal of achieving 50 circular innovations within five years, a target that was introduced together with a new governance structure. Employees from all divisions were encouraged to submit ideas for these new circular innovations, as were outside startups. From the onset, the team of managers that formulated the goal were actively targeting a wide range of R-strategies that included both radical and incremental ideas, and innovation in business models, products, and processes. As such, the goal was intended to be motivational, to spur ideation, and to create alignment by introducing common definitions and measurements within the company. This aligns with the cognitive aspect of sustainability control systems as theorized in Gond et al. (2012): that they facilitate interaction and create opportunities for discussion among people, each bringing to bear different patterns of thinking, mindsets, and practical viewpoints. Other empirical studies have shown that goals and performance indicators are essential in building innovation capability for the very same reasons (Nilsson & Ritzén, 2014).

However, the defined goal at Tools presented several challenges. The first concerned the balance of incremental and radical innovation. In particular, there was an acknowledged need for incremental innovation: i.e., input in the form of ideas that would be relatively easy to achieve within the given timeframe. This was both a question of capacity and impact, since radical innovation requires additional resources and often has less of a chance of yielding commercially viable ideas. In addition, the actual quantified performance in terms of circularity or the impact of innovation was considered too challenging and time-consuming to measure, given the tools and methods available at the time. Instead, if the committee felt

hesitant as to whether all the criteria were met after an initial proof-of-concept, a full lifecycle analysis using external consultants would be part of the evaluation process. No interviewee believed that this approach was perfect. However, despite the many challenges and possible flaws in the approach highlighted by the interviewees, most seemed to agree that this was the most viable way of implementing CE at the time, given the lack of standards and considering how new CE principles were to the firm. Thus, the corporate goal and R-framework were centrally decided acceptable-level decision rules (Cyert & March, 1992).

While no other case has the same quantified corporate goal as Tools, there were additional examples of results controls that were introduced to promote CE principles, such as specific targets for increasing salvage grade in remanufacturing at Trucks, for example. However, comparing results controls for CE with those for other initiatives, it became apparent that the cases here made greater strides in integrating goals such as CO2 reduction, for example, both technically and cognitively (Gond et al., 2012). This difference may be due to several factors: for example, CO2 reduction ambitions were formalized prior to CE ambitions, and well-established controls for monitoring CO2 reduction exist, as well as the fact that this ambition may not have challenged the linear business model in the same way. In all cases, it remained unclear whether, when, and to what extent different circular strategies could complement or hinder other strategic objectives that had more precise goals. This underscores the complexity of managing multiple goals, which may inadvertently favor the status quo if this complexity is not managed effectively. Ethiraj and Levinthal (2009) suggest strategies such as focusing on a subset of goals, dispersing goals across different organizational units, and addressing one goal at a time. However, these approaches also present challenges in terms of incentives and coordination (a challenge already underscored in the previous section).

In summary, using results controls in firms' transformations towards CE is multifaceted. On the one hand, goals and measurements can be used to effectively monitor the accomplishment of ambitions, to emphasize strategic direction, to foster alignment and coordination, and to stimulate radical framings. On the other hand, they can lead to suboptimization and to a focus on aspects that can be quantified or that yield results in the short term, leading to an incremental framing. Both of these contrasting perspectives have been observed in the cases studied, and there are implications stemming from compatibility between CE and existing controls, and how much managerial attention is paid to particular controls. Furthermore, results controls, by their nature, focus on outcomes rather than processes, which can potentially divert attention away from the exploration and learning required to truly achieve those outcomes.

Cultural controls

Several prior studies have highlighted significant challenges related to the cultural aspects within established firms when adopting circular business models (Bertassini et al., 2021). Examples of influencing factors include existing cultural norms, traditional business mindsets, employee attitudes and habits, managerial philosophies, deeply rooted mental schemes, siloed thinking, fear of risks, and internal resistance. In Paper B, it was proposed

that the sheer number of cultural barriers could indicate a substantial challenge for practitioners, which they need guidance in detangling. Furthermore, it is important to note that we need a more nuanced understanding of organizational culture as more than just a barrier to CE implementation. The change management literature (Kotter, 1995) and management control literature (Simons, 1995a) often emphasize organizational culture as a tool that drives organizational change. In the innovation management literature, innovative cultures are understood to be enablers for the exploratory activities that are essential for radical innovation (Menzel et al., 2007). Therefore, following the insights from Paper B, it was considered important to delve deeper into how cultural controls can both facilitate and hinder radical circular innovation in practice.

At Transportation, few interviewees perceived significant challenges to their existing company culture. Instead, some viewed their culture as being positively influenced by experiences in Lean production and manufacturing, where parallels were drawn and synergies found with respect to CE. The philosophical discussion on the Lean idea of "waste" is interesting in comparison to CE. However, in practice, Lean is often reduced to reducing variability or achieving material efficiency in manufacturing (for example, eco-efficiency, cost reduction, waste reduction, and material productivity), as Bocken et al. (2014) discuss. In contrast, from a process management perspective, the same Lean ideas regarding waste may lead to a goal of reducing slack time or skunk work, ultimately fostering a myopic perspective and hindering creativity (Helander et al., 2015) that is detrimental to radical innovation. Therefore, even guiding principles that at first glance may seem to be reasonably in line with CE may need to be examined with respect to whether and how they truly support incremental or radical framing of CE.

At both Tools and Transportation, the shared pursuit of product performance emerged as a challenge to exploring radical circular innovation. Well-established beliefs and ideals about "good performance" in a traditional linear product offerings model posed a dilemma when transitioning to circular value propositions. This quandary was particularly emphasized at Tools and Transportation, as the current differentiation based on the technological performance of combustion engine products was no longer viable. At Transportation, informants also shared examples where colleagues had been pleased with the efforts made to improve the recyclability of their products, and so they questioned the need for additional CE initiatives. This doubt can be attributed to several factors within the realm of cultural controls. Firstly, training and knowledge: i.e., a proper understanding of the full scope of CE. Secondly, propensity or acceptance for risk-taking, as it is easier to judge risks for current business activities than for novel ones, and circular offerings often come with increased complexity. The third factor could include normative management: i.e., who are we and who do we want to be? In a case where the shared core value proposition and belief has always been delivering premium value or the highest-performing products, those values do not necessarily have to be discarded, but they must be understood in the context of CE.

Interviewees reported several instances where experiments and pilot projects successfully shifted attitudes and knowledge regarding CE practices. For instance, Tools initially favored

virgin plastics for handheld tools due to concerns about the quality of recycled materials (for example, resistance to UV light, harsh weather conditions, and use and abuse). However, after experimenting with post-consumer recycled materials in one product line, their experience and competencies with these materials led to a more positive attitude toward them. Another example concerns a radically different business model. Specifically, Tools introduced a B2C sharing solution for consumer goods, which—despite initial uncertainty about its profitability—was frequently praised by senior management and eventually launched internationally in more than eight countries. This experience underscores the importance of senior management's attention and communication as a cultural control to support radical innovation when traditional results controls may be insufficient.

In contrast to the emphasis on organizational culture as a barrier to CE in prior studies, the cultural controls identified in Paper D were the type of managerial control that presented the fewest tensions with CE adoption. Instead, cultural controls emerged as enablers of CE adoption within organizations, playing a crucial role in establishing the legitimacy of CE initiatives. Across all the cases studied, strong senior management commitment to sustainability was considered a consistent and guiding force. This commitment was particularly emphasized in cases facing turbulent business environments, highlighting the pivotal role of organizational culture in driving CE initiatives. As such, it is not necessarily CE itself that needs to be highlighted and communicated, as it was at Tools. Instead, facilitating an organizational culture that emphasizes change and allows for greater risk-taking supports a radical framing of CE.

In sum, prior studies have emphasized cultural aspects at incumbent firms as pertinent barriers to adopting CE practices. The empirical results from this thesis yield a more nuanced perspective that provides further insight into what these cultural aspects entail and how cultural controls can be used to facilitate radical circular innovation.

Closing: Management control systems and framing

In closing, it is suggested that the dominant framing and the MCS used are interlinked factors that influence the adoption of CE principles in organizations. Arguably, since the concept of CE is complex and ambiguous, it is bound to be framed differently by individuals and organizational subunits. Organizations must identify different interdependencies for framings and bring them into light if they wish to foster the adoption of CE principles. In particular, it is suggested that since the MCS is something that can be deliberately designed and changed through interventions, it is therefore a crucial mechanism that can influence individual actors' framing of CE and help the dominant framing (Kaplan, 2008) of CE gain traction within the organization. As the individual cases show, incumbent firms face challenges with legacy MCSs, which are part of the *origin* of particular framings. In addition, when implementing CE, different MCS interventions are being made in terms of action, results, and cultural controls, which act as a mechanism to counter organizational barriers and support organizational transformations towards CE.

A practitioner might therefore ask: how then should we design the MCS? Unfortunately, there is no prescriptive, go-to guide here. CE is an evolving phenomenon, and different types of managerial controls have their own sets of limiting conditions in this regard. In terms of results controls, do we know what the desired result is in terms of CE? Surely one desired result is decoupling resource consumption from economic growth, but how can such a large, systems-level perspective be translated into something that guides daily decisions for product designers? Can we measure those results effectively? In terms of cultural controls, how do we know that the interventions made really influence values and norms? In terms of action controls, do we know what the "desirable actions" are when we are dealing with an evolving phenomenon? There is no easy answer to these questions, but they create fertile ground for interdisciplinary research, as will be discussed in the following section.

6.3 Concluding Discussion and Implications for Research

At many large industrial organizations, such as the ones studied in this thesis, the conventional model of centralized and hierarchical control is increasingly giving way to a more decentralized approach. Responsibility has shifted from formal central control systems to individual managers and team members (Mundy, 2010). However, expecting these individuals to make rational and informed decisions in the complex landscape of CE is a tall order. Decisions made in the early stages of product development, for instance, carry significant implications for a product's circularity potential. These choices often involve intricate trade-offs, dependencies on external stakeholders, and potential economic risks—all against the backdrop of managers' past experiences in a linear logic of production and consumption, on which their career has heretofore been built.

The duality of the two outlined framings of CE as radical or incremental introduces a new layer of tension within organizations, adding to the challenges already inherent to CE implementation. Managerial controls can influence, and are in turn influenced by, whether managers frame CE as a radical or incremental shift. Failing to strike the right balance when using managerial controls can lead to several issues in firms, including slow decision-making, wasted resources, internal instability, and diminished performance. This was observed in practice, as successful pilot projects were forgotten due to a lack of coordination, and the rollout of a CE strategy was postponed due to a lack of clear metrics. Therefore, understanding and thoughtfully applying the appropriate managerial controls becomes pivotal in successfully navigating the intricacy of CE transformation at incumbent firms.

As demonstrated, tensions are especially apparent in the intersection of results controls, resource allocation, and the coordination and integration of different functions. Large organizations are organized in departments that break down complex, interrelated problems and conflicting goals into many simpler problems and goals that make sense within their own specialized, bounded space. At incumbent firms, each of these subunits has its own legacy built on a linear logic. Without effective coordination of CE implementation, subunits may adopt different framings that resonate well within their subunit, thereby complicating the broader transformation of the organization. Furthermore, the tendency to opt for incremental improvements, which require less interdepartmental coordination, risks

diminishing CE to easily achievable goals that do not challenge the existing linear business model.

Researchers, practitioners, and policymakers are increasingly focused on CE targets, goals, and measurements. However, integrating these metrics within broader management control systems is a topic that is still under-explored. As the thesis shows, this is problematic, as there are essential implications to consider. Setting specific targets can catalyze creativity and organizational alignment. However, this may also induce resistance or cause setbacks if there is no agreement on standardized metrics or definitions, undermining CE's legitimacy. Additionally, it is crucial to consider the granularity and timing of these goals. In the rush to meet quantitative performance indicators, teams and organizations might bypass essential steps needed to build the requisite skills and capabilities for long-term CE transformation. This can be particularly detrimental in the context of CE, given the complex nature of the transition away from a linear logic. Moreover, the ease with which certain R-strategies can be measured should not dictate that only these strategies be selected. For instance, strategies such as recycling might offer straightforward, quantifiable metrics, but this ease of measurement should not overshadow strategies such as rethink, which are more challenging to quantify but might offer more significant circularity benefits. This asks how organizations can use MCS in light of complexity and trade-offs.

Towards an integrative framework of radical circular innovation and management control systems

CE inherently entails goals and decisions that are interrelated yet often contradict each other. As argued in this thesis, different framings may intensify organizational tensions and lead to stagnation in CE transformations. This scenario requires managers to be able to manage these resulting tensions. A paradox perspective acknowledges tensions among multiple interrelated, but sometimes conflicting, desirable objectives. From this perspective, paradoxes should not be eliminated or resolved but must be worked through so that they create a more workable certainty in order to enable organizational change (Luscher & Lewis, 2008). This has been extensively discussed in the literature on corporate sustainability (Hahn et al., 2010, 2014, 2018): for example, with respect to the tensions among different sustainability objectives or the tensions and trade-offs between profitability and sustainability (Van der Byl & Slawinski, 2015). Nevertheless, such a perspective is arguably even more relevant to the case of CE, as it entails the same tensions as corporate sustainability and adds in those inherent to the pursuit of different R-strategies. As well, we currently lack the kind of established standards and guidance that we have for sustainability objectives. As such, managing paradoxes at the individual managerial level and at the organizational level becomes relevant for CE transformations in organizations. By identifying and working through inherent paradoxes, organizations have a greater chance of concurrently meeting their strategic objectives for sustainability, profitability, and circularity concurrently. Building on the relationship between MCS and framing argued for in this thesis also creates new avenues for research regarding the role of MCSs and how they can support or impede this way of working.

This thesis observes that managerial controls often are inward-looking and do not favor a more systemic perspective or external actors. This general critique is not new (Lowe & Puxty, 1989), but it is interesting to consider with respect to practitioners' need for partnerships and innovation ecosystem collaborations in order to make CE practices feasible, as well as with respect to the systemic nature of CE in general. As several informants emphasized and as previous studies have noted (e.g., Veleva & Bodkin, 2018), achieving CE often relies on actors external to the firm. Therefore, orchestrating innovation ecosystems and inter-organizational networks have emerged as salient topics in CE research (Hansen & Schmitt, 2021; Konietzko et al., 2020; Parida & Wincent, 2019). Since in a CE, material flows are inherently inter-organizational and inter-sectoral, this is not surprising. However, more research is required to understand inter-organizational learning (Korhonen, Honkasalo, et al., 2018). This is particularly interesting to consider with respect to MCSs since in many cases, the level of analysis for achieving CE performance targets may shift from the individual firm to a more systemic level. Much of the CE literature has focused on the quest for win-win scenarios where several stakeholders are motivated to act because they can identify a potential win. However, going back to the local rationality of subunit goals (Cyert & March, 1992), can we expect a CE to perform well at a systemic level if performance is judged locally by each actor using traditional means based on a linear logic? This may be especially challenging in the context of CE, in light of the lack of standards and consensus on how to assess performance to date. While scholars of CE are increasingly finding best-practice examples of innovation ecosystems, the limitations and assumptions of conventional management controls may need to be considered in conjunction with this.

As argued in Paper D, embracing the foundational principles of the systems thinking intrinsic to CE requires reevaluation and refinement of management controls. Promising alternatives are surfacing, including Langfield-Smith's (2008) study of controls within collaborative alliances and van der Meer-Kooistra and Scapens' (2008) work on control mechanisms within and between organizations, each of which offers noteworthy insights that are arguably transferrable to the context of CE. Given the apparent tensions between traditional MCSs and CE, developing an MCS grounded in adaptive management practices may be imperative. This means implementing management controls while learning which actions are most effective for achieving specified objectives (Eisenhardt & Brown, 1998; Uhl-Bien & Marion, 2009)—an interesting avenue for future research. In addition, further understanding of how complexity theory (Uhl-Bien & Marion, 2009) or systems thinking (Senge, 2006) can be integrated into the design of MCSs is appealing, considering the importance of developing and engaging stakeholders both inside and outside the organization when implementing CE. These emerging perspectives share a unified view, portraying organizations as intricate networks of interrelated, dynamic components and questioning longstanding paradigms of alignment and control. The objective is not consistency among objectives and indicators across organizational units but rather alignment through a continuous iterative process of adjustments among multiple organizational elements (Bellisario et al., 2021).

A perspective that sees organizations as complex systems requires a more reflective and measured approach that acknowledges the limited ability of managers to control or predict outcomes. Thus, the entrenched paradigms of control, determination, and prediction may evolve to encompass notions such as influence, support, and guidance, indicating a shift towards a more nuanced understanding of the managerial role and the use of managerial controls. While numerous MCS tools may retain their utility, their application needs to be studied in this new context (Pavlov & Micheli, 2022).

To summarize this analysis and discussion and to fulfill the aim of this thesis, an integrative framework is proposed to consolidate the analysis and discussion, presenting three distinct modes of organizational behaviors within the context of a CE transition, see Figure 9. This framework, drawing inspiration from Adams et al.'s (2016) framework for sustainability-oriented innovation, delineates three modes of organizational behavior in CE implementation: optimizing, transforming, and systems building. Significantly, it synthesizes the theoretical domains and discoveries of this thesis, ranging from managerial controls to innovation management.

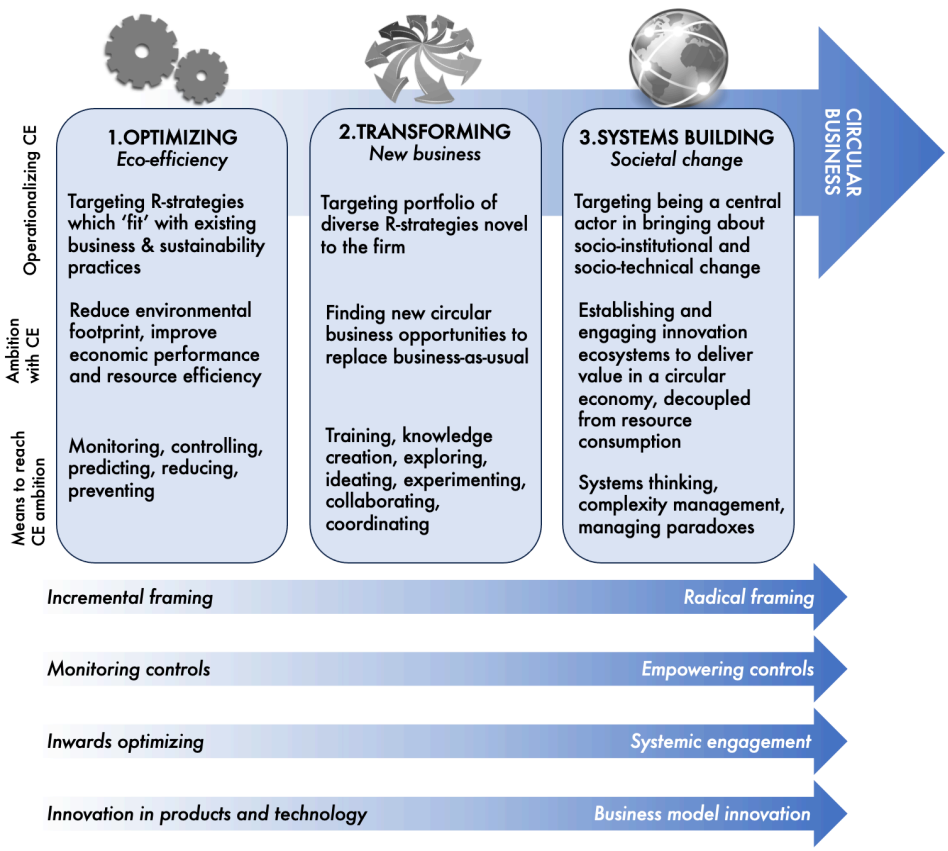


Figure 9: Framework of organizational behaviors in CE transitions.

It is crucial to clarify that the framework does not imply a prescriptive or sequential roadmap. In fact, the observations in this thesis indicate that organizations often concurrently engage in activities characteristic of the different modes. However, based on the analysis here, it should be emphasized that the optimizing behavior is markedly distinct from the others. In particular, it is suggested that solely engaging in the optimizing mode of behavior raises a risk of organizational stagnation, potentially impeding broader transformational shifts towards embracing the principles of CE. Therefore, the framework should be considered as a starting point for scholars to continue finding additional mechanisms that encourage transformational and systems-building behaviors, and for practitioners to understand the implications and requirements of these different modes of behavior. While this thesis primarily uses the theoretical perspectives of exploration and exploitation and MCS, several other relevant perspectives could complement this framework.

Furthermore, this thesis has empirically examined and theoretically substantiated the optimizing and transforming modes, whereas the systems-building mode merits further exploration, as it is a pivotal subject for future research and practitioner needs. Within this framework, systems-building has been shaped predominantly by theorizing on the challenges encountered at the studied firms rather than by examining their actual approaches to systems-building to date. Hence, exploring additional cases that have effectively achieved systems-building behaviors will be crucial to uncovering facilitating factors. The identified theoretical domains of systems thinking, complexity management, and cognitive paradoxes may provide insightful foundations. They serve to rejuvenate research on managerial control and align it more closely with current practitioner needs, particularly those looking to engage with broader innovation ecosystems and to manage conflicting objectives in order to successfully create and deliver value in a CE.

6.4 Final Reflections, Limitations and Ethical Considerations

Limitations of the chosen research methods

There are limitations inherent to the chosen research methods of systematic literature mapping, systematic literature reviews, and interview-based studies. For systematic literature mappings, the data which is analyzed is limited to only the abstract, title, keyword, and journal, which is a very limited set of data if you wish to appreciate the full nature of a study and is highly dependent on the authors' ability to convey this in the abstract, as well as the ability of the mapping researcher to interpret this set of data systematically. Inferences made from mapping studies should, therefore, be made with caution, a limitation that is frequently underscored in Paper A. In this thesis, the mapping database that resulted from Paper A guided the selection of papers for the systematic literature review in Paper B. If a different research team had conducted the literature mapping, they would likely have produced a somewhat different selection of papers and thus identified a somewhat different set barriers. However, the systematic approach applied in Papers A and B is arguably very transparent in outlining each step of the process and providing supplementary open-access data regarding the entire mapping database and analysis.

There are other biases to look out for in the empirical research that formed the basis for Papers C and D, such as participant and researcher biases. In an interview setting, informants are inclined to be friendly and helpful, as people would be in (hopefully) any social setting. This inclination may lead them to provide the answers that they believe the researcher is looking for. This issue was mitigated to some extent by asking open-ended questions to avoid mere agreement or disagreement, and by asking sets of similar but slightly rephrased questions that allowed me to check for coherence.

Another bias that should also be considered is a social acceptability bias, since sustainability and CE are topics with socially acceptable answers. For example, a respondent might be reluctant to share their critique of how sustainability issues are handled within their organization or to admit that they personally do not consider these issues important or relevant. Here, it was important to stress to the participants that the interview data would be handled anonymously. Still, using quotes in qualitative research articles could be problematic in this regard, since colleagues may be able to identify informants by putting pieces of information together. For this reason, the case companies have not been named, the full job titles of the interviewee roles and functions have not been disclosed, and some statements that contained information that could be used to identify the interviewee were not quoted in the appended papers.

Regarding researcher bias, confirmation bias exists when researchers "search" for data that supports their preconceptions, beliefs, or desired results. Instead, all data should be considered in its entirety and each code during analysis treated as important. The inclusive coding process, as suggested by Braun and Clarke (2006), was particularly helpful in this regard. Unexpected codes and themes emerged—an indication that the research team let the data "lead the way" rather than deductively searching for supporting evidence. Creating codebooks and cross-checking each other's coding was also helpful, as it led to discussions among team members and resulted in a more consensual agreement on codes. Another way to mitigate confirmation bias is through informant checks: i.e., going back to the informants to share the analysis. As the research projects were unfolding, we hosted webinars in which we shared the initial analysis with the informants, inviting them to give feedback.

Many of the interviews were conducted during the COVID-19 pandemic, and it was therefore most suitable to hold them using online video conferencing tools to allow for social distancing. However, the implications of this choice have not been the topic of extensive reflection in the appended papers: for example, how this context may have affected empathy building, the exchange of visual cues that support better communication, and the interviewer's ability to assess the workplace environment, which face-to-face meetings would have supported (De Villiers et al., 2022). To mitigate some of the negative aspects of online interviews, I tried to make the environment more relaxed by making small talk at the start of each meeting and using body language. In a handful of cases, the informants provided interviews by telephone (audio-only) or left their cameras off for personal or technical reasons, making it more challenging to create a natural setting. In these few cases,

an in-person interview would likely have yielded more informative responses. Aside from the possible negative aspects of video-conferenced interviews, there were also clear benefits. For example, they were more time-efficient and required less travel to interview informants at geographically dispersed organizations. Because of this, it was also easier to manage the booking of interviews, since they could be done in close succession if schedules allowed but did not necessarily have to take place on the same day or week (which is often desirable when face-to-face interviews require travel).

Choice of publication outlet

It is worth noting that Paper A was published in MDPI's controversial journal, *Sustainability*. Among other things, the academic community has raised questions about this publisher's sheer number of accepted publications and the speed at which the peer-review process takes place, leading to concerns as to whether the process is rigorous enough or if it should be considered a "predatory journal." However, at the time Paper A was submitted, the debate over this publisher had only recently started to emerge. As Paper A highlights, *Sustainability* was also a significant outlet for relevant research, which justified its submission to that outlet as a way to take part in that academic discussion. In retrospect, I wish that this paper had been submitted elsewhere, since it is unfortunate to be associated with questionable research outlets when you are proud of the research that you did.

Consequently, I wrote an essay on predatory journals and my own experience as part of the KTH doctoral course "The Sustainable Scientist." I received feedback that the course administrator would include this as a topic for future offerings, so hopefully my contribution there will help future doctoral students to be more informed regarding the existence of predatory journals and questionable publication practices.

Limitations of the meta-analysis in this thesis

This comprehensive summary adds new perspectives to the appended papers in the form of a meta-analysis. At first, I selected the concept of cognitive framing as an experiment to provide structure to the analysis. However, after experimenting with it, I found it had interesting implications that very well encapsulated the patterns I observed during the empirical investigation. Nevertheless, this analysis should be viewed as indicative rather than decisive and as offering an invitation for new studies. Specifically, it would be interesting to conduct empirical research that makes clearer distinctions between the individual, team, and organizational levels of framing.

In addition, the cases here only shed light on a handful of firms at a particular point in time when the implementation of CE was relatively new. They therefore offer only a snapshot of what the firms were doing and what they had done. The temporal dimension behind the origins and consequences of each framing relies on abductive reasoning. In other words, the observations are incomplete, and plausible inferences must be made by linking pieces of evidence together and theorizing. In contrast, a longitudinal study would have more potential for understanding the evolution of managerial controls, and a longer data collection time horizon would have provided more opportunities to observe performance in

terms of radical circular innovation as an outcome. Additionally, the origins and consequences of the two framings could be investigated on several different levels. For example, the origins discussed in this thesis are at an organizational level and concern the contexts in which individuals find themselves. In contrast, a different research design could have investigated this on an individual level (for example, in terms of job tenure, individual goals, views on sustainability, and so on).

Reflections on the overall research approach and choice of theory

A pragmatist approach invites a wide selection of theories and concepts. At the start of the doctoral studies from which this thesis emerged, I had freedom within the project to pursue several different paths. Having many different options is both liberating and burdensome for a junior scholar studying an evolving phenomenon. While this comprehensive summary outlines my research journey as a form of stepwise "funneling" or a single nice "red thread," the true picture is, of course, much messier. Being led by the needs I observed in the industry and guided by the evolving CE literature led to a greater understanding of the phenomena as a whole. However, a narrower scope would likely have yielded more detailed insights on specific topics, reflecting the necessary tradeoff between depth and breadth.

In terms of theories and concepts, the thesis includes several concepts, such as the R-strategy framework, exploration vs. exploitation, management controls, innovation, ambidexterity, and so on. Such concepts allow researchers (and practitioners) to discuss and build on ideas (Morgan, 1980). However, the concepts chosen can also be limiting and have important implications. For example, the R-strategy framework does little justice to topics such as the role of energy or biodegradable materials in CE. Nevertheless, I have seen it spark fruitful discussions among business developers and product designers. Thus, while this particular choice may not have significant implications for the thesis, it may have effects when interreacting with practitioners. Likewise, there are several alternatives to Merchant and van der Stede's categorization of management controls (2017), such as that of Simons (1995a). Alternative schemes may be more relevant for understanding the balance between innovation and control, but I personally found them less intuitive and relatable for practitioners. Regarding how the different framings I present in this comprehensive summary conceptualize and relate to innovation, radical versus incremental only represents one dimension of innovation. In contrast, other distinctions, such as between architectural and disruptive innovation as in Seebode et al. (2012), may offer other nuances. Therefore, researchers such as myself must strike a balance between understanding the "big picture" and practical implications of zooming in, a balance that is especially important when interacting with practitioners, since it may have implications for real-world decisions.

On the same note, with respect to ethical considerations, circular does not necessarily mean sustainable. We still need a better understanding of many aspects of the relationship between sustainability and CE (Corvellec et al., 2021; Kirchherr & van Santen, 2019). Personally, I share the view of Geissdoerfer et al. (2017), in that the relationship between CE and sustainability is that of a subset. In other words, CE is one among several solutions for

fostering sustainable systems. It is not a panacea but instead presents solutions to actual problems (Corvellec et al., 2021).

Furthermore, one of the objectives of this thesis is to help guide large industrial companies in facilitating radical circular innovation. However, the thesis does not try to judge their performance in terms of sustainability, something that many large companies are struggling with currently. The focus on *radical circular innovation* here is relative, since it judges novelty in relation to traditional linear production and consumption practices. While my definition shared earlier says nothing about the aims of sustainable development, it does acknowledge the substantial evidence that current practices are not sustainable and must undergo significant changes and that CE presents promising alternatives if not reduced to only incremental improvements.

Although they are outside of the scope of this thesis, promising developments are unfolding in terms of standards, taxonomies, sustainability reporting, and CE indicators—something that industrial practitioners are in dire need of. Remarkably, recent studies by Opferkuch et al. (2021, 2022) have found that most firms—even those with lofty CE ambitions—report their CE performance in terms of waste management operations and lower-ranking R-strategies (such as recycling or recovery). Managing end-of-pipe waste does not belong in a CE paradigm, but is the final phase of the linear economy (Stahel, 2019). If these are the results controls that firms use to measure performance, their CE transformation cannot be expected to be complete. Investigating how companies adapt to these forthcoming standards, their implications for managerial controls, and how they facilitate or impede radical circular innovation is an interesting avenue for future research.

7. Conclusions and Implications for Practice

Planet Earth can be described as a closed system with regard to matter. Of course there are exceptions. We are hit by asteroids and space dust falling every day, and we send out spaceships, satellites, and the occasional electric car for unforetold reasons. However, these are insignificant when compared to the total amount of matter in the system. With a growing population and rising standards of living, continuing our "take-make-waste" logic of production and consumption cannot be sustained. Instead, the idea of a circular economy has been proposed, in which the utility of products and the purity of materials is maintained to the greatest extent possible, for as long as possible. This places fundamentally new requirements on industrial firms.

By synthesizing extant research on the barriers to CE through a systematic literature mapping, a subsequent systematic literature review, and a series of empirical case studies, this thesis contributes to the rapidly developing literature stream on barriers to CE (RQ1). Building on the identified challenges, it incorporates the perspectives of innovation management (RQ2), and in particular how management control systems can support radical circular innovation at large industrial incumbent firms (RQ3). It argues that organizations need to consider how radical innovation can be facilitated to achieve the changes that full

adoption of CE principles requires. This is particularly important for large, incumbent industrial firms, where radical innovation may be especially challenging.

The thesis suggests that individual managers interpret (*frame*) CE following two different patterns. The differences between these two framings have significant implications for how organizations set goals, measure success, and navigate the challenges of implementing CE principles, and the coexistence of both framings creates tensions and conflicts within the organization. As a whole, building upon the challenges identified in the case studies, the analysis and integrative framework in this thesis shed light on RQ2 and RQ3, as will be elaborated on below.

In particular, it is argued that the consequences and conflicts resulting from an incremental framing may be detrimental to achieving the changes on which radical circular innovation relies. Therefore, it is crucial to understand exactly why these two different framings emerge (i.e. their origin), how they gain traction (i.e. become the dominant framing), as well as their consequences. In light of this, the thesis investigates the use of management control systems (MCS), seeking to identify potential linear lock-ins and incompatibilities between CE and existing MCSs, as well as how MCSs can be crucial mechanisms to support CE transformations (RQ3). It finds that managerial controls are crucial mechanisms to influence how and which dominant framing prevails, building on empirical observations on how radical circular innovation is facilitated (RQ2) and theorizing around the relationship between cognitive framing and MCS. In practice, this means that managers can influence the effectiveness of CE implementation through managerial controls, potentially alleviating the tensions and conflicts that arise from the clash between contrasting framings within the same organization.

The use of results controls (e.g., goals and measurements) has important implications for practitioners. First, firms need to consider how much employees and activities are guided by results controls based on a linear logic (e.g., manufacturing quantities, sales volumes, or assembly time) and contrast these to ones based on a circular logic (e.g., remanufacturing quantities, product utilization rates, or disassembly time). Second, results controls must balance short-term performance with long-term ambitions. This is particularly important in relation both to incremental and radical innovation and to ambitions in sustainability and CE. Third, the use of results controls to measure CE implementation must be carefully considered so that they are able to both monitor performance and promote ideation in pursuit of a variety of R-strategies, rather than emphasizing the ones that are most convenient to measure or that achieve the best short-term performance. Lastly, the amount of managerial attention to results controls must be considered. This refers to the finding that CE ambitions are, in practice, compared against other sustainability ambitions, and may receive less attention than those with a better fit with an organization's dominant controls.

The thesis found that even in instances that pointed to conflicting results controls, the studied cases emphasized the role of cultural controls as guidance for CE implementation. They are, therefore, pivotal enablers of CE adoption within organizations and are crucial for

establishing the legitimacy of CE initiatives and alleviating conflicts among controls. In particular, cultural controls do not necessarily concern CE initiatives directly. Instead, they can facilitate empowerment and emphasize the need for organizational transformation and for assuming a role in a larger sustainability transition. This creates the necessary conditions for facilitating radical circular innovation.

Synthesizing these insights yields an integrative framework for classifying firm behaviors in CE transitions as either *optimizing*, *transforming*, or *systems building*. Each of these necessitates a distinct managerial approach to innovation and management controls (RQ2 and RQ3). This observation paves the way for future exploration and practical use at incumbent firms pursuing CE transformations. In particular, it is argued that managerial practices at incumbent firms must go beyond optimizing behavior if they are to effectively support CE transitions. This research thereby constitutes a stepping-stone for understanding and navigating the multifaceted landscapes of organizational change, innovation, and management controls in the pursuit of a system where production and consumption put less stress on finite natural resources.

■

References

Adams, R., Jeanrenaud, S., Bessant, J., Denyer, D., & Overy, P. (2016). Sustainability-oriented innovation: A systematic review. *International Journal of Management Reviews*, *18*(2), 180–205. https://doi.org/10.1111/ijmr.12068

Ahrens, T., & Chapman, C. S. (2004). Accounting for flexibility and efficiency: A field study of management control systems in a restaurant chain*. *Contemporary Accounting Research*, *21*(2), 271–301. https://doi.org/10.1506/VJR6-RP75-7GUX-XHoX

Ahuja, G., & Morris Lampert, C. (2001). Entrepreneurship in the large corporation: A longitudinal study of how established firms create breakthrough inventions. *Strategic Management Journal*, *22*(6–7), 521–543. https://doi.org/10.1002/smj.176

Andriopoulos, C., & Lewis, M. W. (2009). Exploitation-exploration tensions and organizational ambidexterity: Managing paradoxes of innovation. *Organization Science*, *20*(4), 696–717.

Anthony, R. N., & Govindarajan, V. (2006). *Management Control Systems* (12th ed.). McGraw-Hill.

Asif, F. M. A., Roci, M., Lieder, M., Rashid, A., Štimulak, M., Halvordsson, E., & De Bruijckere, R. (2018). A practical ICT framework for transition to circular manufacturing systems. *Procedia CIRP*, *72*, 598–602. https://doi.org/10.1016/j.procir.2018.03.311

Baker, M., & Schaltegger, S. (2015). Pragmatism and new directions in social and environmental accountability research. *Accounting, Auditing & Accountability Journal*, *28*(2), 263–294. https://doi.org/10.1108/AAAJ-08-2012-01079

Barnes, J. H. (1984). Cognitive biases and their impact on strategic planning. *Strategic Management Journal*, *5*(2), 129–137.

Baumgartner, R. J., & Ebner, D. (2010). Corporate sustainability strategies: Sustainability profiles and maturity levels. *Sustainable Development*, *18*(2), 76–89. https://doi.org/10.1002/sd.447

Belk, R. (2014). You are what you can access: Sharing and collaborative consumption online. *Journal of Business Research*, *67*(8), 1595–1600. https://doi.org/10.1016/j.jbusres.2013.10.001

Bellisario, A., Pavlov, A., & van, der S. M. P. (2021). The role of performance measurement in aligning operations with strategy: Sustaining cognitive processes of internal alignment. *International Journal of Operations & Production Management*, *41*(12), 1879–1907. https://doi.org/10.1108/IJOPM-02-2021-0081

Berry, A. J., Coad, A. F., Harris, E. P., Otley, D. T., & Stringer, C. (2009). Emerging themes in management control: A review of recent literature. *The British Accounting Review*, *41*(1), 2–20. https://doi.org/10.1016/j.bar.2008.09.001

Bertassini, A. C., Ometto, A. R., Severengiz, S., & Gerolamo, M. C. (2021). Circular economy and sustainability: The role of organizational behaviour in the transition journey. *Business Strategy and the Environment*, *30*(7), 3160–3193. https://doi.org/10.1002/bse.2796

Bessant, J., Öberg, C., & Trifilova, A. (2014). Framing problems in radical innovation. *Industrial Marketing Management*, *43*(8), 1284–1292. https://doi.org/10.1016/j.indmarman.2014.09.003

Bocken, N. M. P., Short, S. W., Rana, P., & Evans, S. (2014). A literature and practice review to develop sustainable business model archetypes. *Journal of Cleaner Production*, *65*, 42–56. https://doi.org/10.1016/j.jclepro.2013.11.039

Bocken, N., & Ritala, P. (2021). Six ways to build circular business models. *Journal of Business Strategy*. https://doi.org/10.1108/JBS-11-2020-0258

Boons, F., & Lüdeke-Freund, F. (2013). Business models for sustainable innovation: State-of-the-art and steps towards a research agenda. *Journal of Cleaner Production*, *45*, 9–19. https://doi.org/10.1016/j.jclepro.2012.07.007

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Braungart, W. M. an M. (1998). The NEXT industrial revolution. *The Atlantic*. https://www.theatlantic.com/magazine/archive/1998/10/the-next-industrial-revolution/304695/

Briguglio, M., Llorente-González, L. J., Meilak, C., Pereira, Á., Spiteri, J., & Vence, X. (2021). Born or grown: Enablers and barriers to circular business in Europe. *Sustainability*, *13*(24), Article 24. https://doi.org/10.3390/su132413670

Calvin, K., Dasgupta, D., Krinner, G., Mukherji, A., Thorne, P. W., Trisos, C., Romero, J., Aldunce, P., Barrett, K., Blanco, G., Cheung, W. W. L., Connors, S., Denton, F., Diongue-Niang, A., Dodman, D., Garschagen, M., Geden, O., Hayward, B., Jones, C., … Péan, C. (2023). *IPCC 2023: Climate Change 2023: Synthesis Report* (First). Intergovernmental Panel on Climate Change (IPCC). https://doi.org/10.59327/IPCC/AR6-9789291691647

Cassell, C., & Symon, G. (2004). *Essential guide to qualitative research methods in organizational research*. SAGE Publications.

Chenhall, R. H. (2003). Management control systems design within its organizational context: Findings from contingency-based research and directions for the future. *Accounting, Organizations and Society*, *28*(2–3), 127–168. https://doi.org/10.1016/S0361-3682(01)00027-7

Cherryholmes, C. H. (1992). Notes on pragmatism and scientific realism. *Educational Researcher*, *21*(6), 13–17. https://doi.org/10.3102/0013189X021006013

Chiesa, V., Frattini, F., Lamberti, L., & Noci, G. (2009). Exploring management control in radical innovation projects. *European Journal of Innovation Management*, *12*(4), 416–443. https://doi.org/10.1108/14601060910996909

Clarke, S. F., & Roome, N. J. (1995). Managing for environmentally sensitive technology: Networks for collaboration and learning. *Technology Analysis & Strategic Management*, *7*(2), 191–216. https://doi.org/10.1080/09537329508524204

Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Aministrative Science Quarterly, 35*(1), 128–152.

Committee on Science Engineering and Public Policy. (2009). *On being a scientist: A guide to responsible conduct in research*. The National Academies. https://doi.org/10.1093/actrade/9780192804662.003.0006

Corvellec, H., Stowell, A. F., & Johansson, N. (2021). Critiques of the circular economy. *Journal of Industrial Ecology 26*(2), 1–12. https://doi.org/10.1111/jiec.13187

Creswell, J. W. (2014). *Research design: Qualitative, quantitative, and mixed methods approaches* (4th ed.). SAGE Publications, Inc.

Crossan, M. M., & Apaydin, M. (2010). A Multi-dimensional framework of organizational innovation: A systematic review of the literature. *Journal of Management Studies*, *47*(6), 1154–1191. https://doi.org/10.1111/j.1467-6486.2009.00880.x

Crutzen, N., Zvezdov, D., & Schaltegger, S. (2017). Sustainability and management control: Exploring and theorizing control patterns in large European firms. *Journal of Cleaner Production*, *143*, 1291–1301. https://doi.org/10.1016/j.jclepro.2016.11.135

Cyert, R. M., & March, J. G. (1992). *A Behavioral Theory of the Firm*. Blackwell Publishers Inc.

Dahlsrud, A. (2008). How corporate social responsibility is defined: An analysis of 37 definitions. *Corporate Social Responsibility and Environmental Management*, *15*(1), 1–13. https://doi.org/10.1002/csr.132

De Villiers, C., Farooq, M. B., & Molinari, M. (2022). Qualitative research interviews using online video technology – challenges and opportunities. *Meditari Accountancy Research*, *30*(6), 1764–1782. https://doi.org/10.1108/MEDAR-03-2021-1252

DeCuir-Gunby, J. T., Marshall, P. L., & McCulloch, A. W. (2011). Developing and using a codebook for the analysis of interview data: An example from a professional development research project. *Field Methods*, *23*(2), 136–155. https://doi.org/10.1177/1525822X10388468

den Hollander, M. C., Bakker, C. A., & Hultink, E. J. (2017). Product design in a circular economy: Development of a typology of key concepts and terms. *Journal of Industrial Ecology*, *21*(3), 517–525. https://doi.org/10.1111/jiec.12610

Denyer, D., & Tranfield, D. (2009). Producing a systematic review. In *The Sage Handbook of Organizational Research Methods* (pp. 671–689).

Dewar, R. D., & Dutton, J. E. (1986). The adoption of radical and incremental innovations: An empirical analysis. *Management Science*, *32*(11), 1422–1433. https://doi.org/10.1287/mnsc.32.11.1422

Diaz Lopez, F. J., Bastein, T., & Tukker, A. (2019). Business model innovation for resource-efficiency, circularity and cleaner production: What 143 cases tell us. *Ecological Economics*, *155* , 20–35. https://doi.org/10.1016/j.ecolecon.2018.03.009

Dyllick, T., & Hockerts, K. (2017). Beyond the business case for corporate sustainability. *Corporate Environmental Responsibility*, *141*, 213–224. https://doi.org/10.4324/9781315259277-7

Eisenhardt, K. M. (1985). Control: Organizational and economic approaches. *Management Science*, *31*(2), 134–149.

Eisenhardt, K. M. (1989). Building theories from case study research. *Academy of Management Review*, *14*(4), 532–550. https://doi.org/10.5465/amr.1989.4308385

Eisenhardt, K. M., & Brown, S. L. (1998). Competing on the edge: Strategy as structured chaos. *Long Range Planning*, *31*(5), 786–789.

Eisenhardt, K. M., & Tabrizi, B. N. (1995). Accelerating adaptive processes: Product innovation in the global computer industry. *Administrative Science Quarterly 40*(1), 84–110.

Ellen MacArthur Foundation. (2013). *Towards the Circular Economy: Economic and business rationale for accelerated transition*. Report.

Erkman, S. (1997). Industrial ecology: An historical view. *Journal of Cleaner Produdction 5*(2), 1–2.

Ethiraj, S. K., & Levinthal, D. A. (2009). Hoping for A to Z while rewarding only A: Complex organizations and multiple goals. *Organization Science*, *20*(1), 4–21. https://doi.org/10.1287/orsc.1080.0358

Fink, A. (2014). *Conducting research literature reviews: From the Internet to paper* (4th ed.). *SAGE Publications*.

Försterling, G., Orth, & Gellert, B. (2023). Transition to a circular economy in Europe through new business models: Barriers, drivers, and policy making. *Sustainability*, *15*, 8212. https://doi.org/10.3390/su15108212

Friedman, M. (1970, September 13). A Friedman doctrine:The social responsibility of business is to increase its profits. *The New York Times*, 17.

Frishammar, J., & Parida, V. (2019). Circular business model transformation: A roadmap for incumbent firms. *California Management Review*, *61*(2), 5–29. https://doi.org/10.1177/0008125618811926

Gassmann, O., Widenmayer, B., & Zeschky, M. (2012). Implementing radical innovation in the business: The role of transition modes in large firms. *R&D Management*, *42*(2), 120–132. https://doi.org/10.1111/j.1467-9310.2011.00670.x

Geels, F. W. (2004). From sectoral systems of innovation to socio-technical systems. *Research Policy*, *33*(6–7), 897–920. https://doi.org/10.1016/j.respol.2004.01.015

Geissdoerfer, M., Pieroni, M. P. P., Pigosso, D. C. A., & Soufani, K. (2020). Circular business models: A review. *Journal of Cleaner Production*, *277*, 123741. https://doi.org/10.1016/j.jclepro.2020.123741

Geissdoerfer, M., Savaget, P., Bocken, N. M. P., & Hultink, E. J. (2017). The circular economy – A new sustainability paradigm? *Journal of Cleaner Production*, *143*, 757–768. https://doi.org/10.1016/j.jclepro.2016.12.048

Ghisellini, P., Cialani, C., & Ulgiati, S. (2016). A review on circular economy: The expected transition to a balanced interplay of environmental and economic systems. *Journal of Cleaner Production*, *114*, 11–32. https://doi.org/10.1016/j.jclepro.2015.09.007

Gibson, C. B., & Birkinshaw, J. (2004). The antecedents, consequences, and mediating role of organizational ambidexterity. *The Academy of Management Journal*, *47*(2), 209–226.

Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013). Seeking qualitative rigor in inductive research: Notes on the Gioia methodology. *Organizational Research Methods*, *16*(1), 15–31. https://doi.org/10.1177/1094428112452151

Gond, J.-P., Grubnic, S., Herzig, C., & Moon, J. (2012). Configuring management control systems: Theorizing the integration of strategy and sustainability. *Management Accounting Research*, *23*(3), 205–223. https://doi.org/10.1016/j.mar.2012.06.003

Greenwood, R., & Hinings, C. R. (1993). Understanding strategic change: The contribution of archetypes. *Academy of Management Journal*, *36*(5), 1052–1081. https://doi.org/10.2307/256645

Guba, E. G., & Lincoln, Y. A. S. (1994). Competing paradigms in qualitative research. In Denzin, N. K., & Lincoln, Y. S. (Eds.), *Handbook of qualitative research*. Sage Publications, Inc.

Guldmann, E., & Huulgaard, R. D. (2020). Barriers to circular business model innovation: A multiple-case study. *Journal of Cleaner Production*, *243*, 118160. https://doi.org/10.1016/j.jclepro.2019.118160

Haffar, M., & Searcy, C. (2019). How organizational logics shape trade-off decision-making in sustainability. *Long Range Planning*, *52*(6), 101912. https://doi.org/10.1016/j.lrp.2019.101912

Hahn, T., Figge, F., Pinkse, J., & Preuss, L. (2010). Trade-offs in corporate sustainability: You can't have your cake and eat it. *Business Strategy and the Environment*, *19*(4), 217–229. https://doi.org/10.1002/bse.674

Hahn, T., Figge, F., Pinkse, J., & Preuss, L. (2018). A Paradox Perspective on Corporate Sustainability: Descriptive, Instrumental, and Normative Aspects. *Journal of Business Ethics*, *148*, 235–248. https://doi.org/10.1007/s10551-017-3587-2

Hahn, T., Preuss, L., Pinkse, J., & Figge, F. (2014). Cognitive frames in corporate sustainability: Managerial sensemaking with paradoxical and business case frames. *The Academy of Management Review*, *39*, 463–487. https://doi.org/10.5465/amr.2012.0341

Hall, J., & Vredenburg, H. (2003). The challenges of innovating for sustainable development. *MIT Sloan Management Review*, *45*(1), 61–68.

Hansen, E. G., & Große-Dunker, F. (2013). Sustainability-oriented innovation. In S. O. Idowu, N. Capaldi, L. Zu, & A. D. Gupta (Eds.), *Encyclopedia of corporate social responsibility* (pp. 2407–2417). Springer. https://doi.org/10.1007/978-3-642-28036-8_552

Hansen, E. G., & Schaltegger, S. (2016). The sustainability balanced scorecard: A systematic review of architectures. *Journal of Business Ethics*, *133*(2), 193–221. https://doi.org/10.1007/s10551-014-2340-3

Hansen, E. G., & Schmitt, J. C. (2021). Orchestrating cradle-to-cradle innovation across the value chain: Overcoming barriers through innovation communities, collaboration mechanisms, and intermediation. *Journal of Industrial Ecology*, *25*(3), 627–647. https://doi.org/10.1111/jiec.13081

Hansen, E. G., Wicki, S., & Schaltegger, S. (2018). *Loosing thrust in radical greentech innovation: A case study on failed ambidexterity ,anagement*. Unpublished paper. https://doi.org/10.13140/RG.2.2.13987.07209

Hart, S. L. (1997). Beyond greening: Strategies for a sustainable world. *Harvard Business Review*, *75*(1), 66–76.

He, Z.-L., & Wong, P.-K. (2004). Exploration vs. exploitation: An empirical test of the ambidexterity hypothesis. *Organization Science*, *15*(4), 481–494. https://doi.org/10.1287/orsc.1040.0078

Helander, M., Bergqvist, R., Stetler, K. L., & Magnusson, M. (2015). Applying lean in product development-enabler or inhibitor of creativity? *International Journal of Technology Management*, *68*(1–2), 49–69. https://doi.org/10.1504/IJTM.2015.068774

Hockerts, K. (2015). A cognitive perspective on the business case for corporate sustainability. *Business Strategy and the Environment*, *24*(2), 102–122. https://doi.org/10.1002/bse.1813

Hofmann, F., & Jaeger-Erben, M. (2020). Organizational transition management of circular business model innovations. *Business Strategy and the Environment*, *August*. https://doi.org/10.1002/bse.2542

IRP. (2019). *Global resources outlook 2019: Natural resources for the future we want*. UN. https://doi.org/10.18356/689a1a17-en

James, K. L., Randall, N. P., & Haddaway, N. R. (2016). A methodology for systematic mapping in environmental sciences. *Environmental Evidence*, *5*(1), 1–13. https://doi.org/10.1186/s13750-016-0059-6

Jensen, M. C. (2002). Value maximization, stakeholder theory, and the corporate objective function. *Business Ethics Quarterly 12*,(2), 235–256. https://www.jstor.org/stable/3857812

Kaplan, S. (2008). Framing contests: Strategy making under uncertainty. *Organization Science*, *19*(5), 729–752. https://doi.org/10.1287/orsc.1070.0340

Kennedy, S., Whiteman, G., & van den Ende, J. (2017). Radical innovation for sustainability: The power of strategy and open innovation. *Long Range Planning*, *50*(6), 712–725. https://doi.org/10.1016/j.lrp.2016.05.004

Kirchherr, J., Reike, D., & Hekkert, M. (2017). Conceptualizing the circular economy: An analysis of 114 definitions. *Resources, Conservation and Recycling*. https://doi.org/10.1016/j.resconrec.2017.09.005

Kirchherr, J., & van Santen, R. (2019). Research on the circular economy: A critique of the field. *Resources, Conservation and Recycling*, *151*(August), 104480. https://doi.org/10.1016/j.resconrec.2019.104480

Kitto, S. C., Chesters, J., & Grbich, C. (2008). Quality in qualitative research: Criteria for authors and assessors in the submission and assessment of qualitative research articles for the Medical Journal of Australia. *Medical Journal of Australia*, *188*(4), 243–246. https://doi.org/10.5694/j.1326-5377.2008.tb01595.x

Konietzko, J., Bocken, N., & Hultink, E. J. (2020). Circular ecosystem innovation: An initial set of principles. *Journal of Cleaner Production*, *253*, 119942. https://doi.org/10.1016/j.jclepro.2019.119942

Korhonen, J., Honkasalo, A., & Seppälä, J. (2018). Circular economy: The concept and its limitations. *Ecological Economics*, *143*, 37–46. https://doi.org/10.1016/j.ecolecon.2017.06.041

Korhonen, J., Nuur, C., Feldmann, A., & Birkie, S. E. (2018). Circular economy as an essentially contested concept. *Journal of Cleaner Production*, *175*, 544–552. https://doi.org/10.1016/j.jclepro.2017.12.111

Kotter, J. P. (1995). Leading change: Why transformation efforts fail. *Harvard Business Review Magazine* (*March-April)*.

Langfield-Smith, K. (1997). Management control systems and strategy: A critical review. *Accounting, Organizations and Society*, *22*(2), 207–232. https://doi.org/10.1016/S0361-3682(95)00040-2

Langfield-Smith, K. (2008). The relations between transactional characteristics, trust and risk in the start-up phase of a collaborative alliance. *Management Accounting Research*, *19*(4), 344–364. https://doi.org/10.1016/j.mar.2008.09.001

Lavie, D., Stettner, U., & Tushman, M. L. (2010). Exploration and exploitation within and across organizations. *Academy of Management Annals*, *4*(1), 109–155. https://doi.org/10.1080/19416521003691287

Leonard-Barton, D. (1992). Core capabilities and core rigidities: A paradox in managing new product development. *Strategic Management Journal 13*, 111–125. https://doi.org/10.1142/9789814295505_0002

Levinthal, D. A., & March, J. G. (1993). The myopia of learning. *Strategic Management Journal*, *14*(May), 95–112. https://doi.org/10.1002/smj.4250141009

Levitt, B., & March, J. G. (1988). Organizational Learning. *Annual Review of Sociology*, *14*(1), 319–338. https://doi.org/10.1146/annurev.so.14.080188.001535

Lewandowski, M. (2016). Designing the business models for circular economy—Towards the conceptual framework. *Sustainability*, *8*(1), 43. https://doi.org/10.3390/su8010043

Lieder, M., & Rashid, A. (2016). Towards circular economy implementation: A comprehensive review in context of manufacturing industry. *Journal of Cleaner Production*, *115*, 36–51. https://doi.org/10.1016/j.jclepro.2015.12.042

London, B. (1932). *Ending the depression through planned obsolescence*. https://hdl.handle.net/2027/wu.89097035273

Lowe, T., & Puxty, T. (1989). The problems of a paradigm: A critique of the prevailing orthodoxy in management control. In W. F. Chua, T. Lowe, & T. Puxty (Eds.), *Critical perspectives in management control* (pp. 9–26). Palgrave Macmillan UK. https://doi.org/10.1007/978-1-349-07658-1_2

Luscher, L., & Lewis, M. (2008). Organizational change and managerial sensemaking: Working through paradox. *Academy of Management Journal*, *51*, 221–240. https://doi.org/10.5465/AMJ.2008.31767217

March, J. G. (1991). Exploration and exploitation in organizational learning. *Organizational Science*, *2*(1), 71–87.

Mathieux, F., Ardente, F., Bobba, S., Nuss, P., Blengini, G., Alves, D. P., Blagoeva, D., Torres, D. M. C., Wittmer, D., Pavel, C., Hamor, T., Saveyn, H., Gawlik, B., Orveillon, G., Huygens, D., Garbarino, E., Tzimas, E., Bouraoui, F., & Solar, S. (2018, January 5). *Critical raw materials and the circular economy – Background report*. JRC Publications Repository. https://doi.org/10.2760/378123

McDonough, W., & Braungart, M. (2002). *Cradle to cradle: Remarking the way we make things*. North Point Press.

McGrath, R. G. (2001). Exploratory learning, innovative capacity, and managerial oversight. *Academy of Management Journal*, *44*(1), 118–131. https://doi.org/10.2307/3069340

Meadows, D. H., Randers, J., & Meadows, D. L. (2006). *The limits to growth: The 30-year update* (Reprint). Earthscan.

Menzel, H. C., Aaltio, I., & Ulijn, J. M. (2007). On the way to creativity: Engineers as intrapreneurs in organizations. *Technovation 27*, 732–743. https://doi.org/10.1016/j.technovation.2007.05.004

Merchant, K. A., & Van der Stede, W. A. (2017). *Management control systems: Performance measurement, evaluation and incentives* (4th ed.). Pearson.

Miles, M., Huberman, M., & Saldaña, O. (2014). *Qualitative data analysis: A methods sourcebook*. SAGE Publications Inc.

Morgan, G. (1980). Paradigms, metaphors, and puzzle solving in organization theory. *Administrative Science Quarterly*, *25*(4), 605–622.

Müller-Stewens, B., Widener, S. K., Möller, K., & Steinmann, J.-C. (2020). The role of diagnostic and interactive control uses in innovation. *Accounting, Organizations and Society*, *80*, 101078. https://doi.org/10.1016/j.aos.2019.101078

Mundy, J. (2010). Creating dynamic tensions through a balanced use of management control systems. *Accounting, Organizations and Society*, *35*(5), 499–523. https://doi.org/10.1016/j.aos.2009.10.005

Nadkarni, S., & Barr, P. S. (2008). Environmental context, managerial cognition, and strategic action: An integrated view. *Strategic Management Journal*, *29*(13), 1395–1427. https://doi.org/10.1002/smj.717

Nilsson, S., & Ritzén, S. (2014). Exploring the use of innovation performance measurement to build innovation capability in a medical device company. *Creativity and Innovation Management*, *23*(2), 183–198. https://doi.org/10.1111/caim.12054

Nußholz, J. L. K. (2017). Circular business models: Defining a concept and framing an emerging research field. *Sustainability (Switzerland)*, *9*(10), 14–17. https://doi.org/10.3390/su9101810

Okoli, C., & Schabram, K. (2011). A Guide to conducting a systematic literature review of information systems research. Working paper. https://doi.org/10.2139/ssrn.1954824

Opferkuch, K., Caeiro, S., Salomone, R., & Ramos, T. B. (2021). Circular economy in corporate sustainability reporting: A review of organisational approaches. *Business Strategy and the Environment*, *30*(8), 4015–4036. https://doi.org/10.1002/bse.2854

Opferkuch, K., Caeiro, S., Salomone, R., & Ramos, T. B. (2022). Circular economy disclosure in corporate sustainability reports: The case of European companies in sustainability rankings. *Sustainable Production and Consumption 32*. 436-456. https://doi.org/10.1016/j.spc.2022.05.003

O'Reilly, C. A., & Tushman, M. L. (2013). Organizational ambidexterity: Past, present and future. *Academy of Management Perspectives, 27*(4) 324–338. https://doi.org/10.5465/amp.2013.0025

Parida, V., & Wincent, J. (2019). Why and how to compete through sustainability: A review and outline of trends influencing firm and network-level transformation. *International Entrepreneurship and Management Journal*, *15*(1), 1–19. https://doi.org/10.1007/s11365-019-00558-9

Patton, M. Q. (2002). *Qualitative research and evaluation methods* (3rd ed). Sage Publications.

Pauliuk, S. (2018). Critical appraisal of the circular economy standard BS 8001:2017 and a dashboard of quantitative system indicators for its implementation in organizations. *Resources, Conservation and Recycling*, *129*, 81–92. https://doi.org/10.1016/j.resconrec.2017.10.019

Pavlov, A., & Micheli, P. (2022). Rethinking organizational performance management: A complexity theory perspective. *International Journal of Operations & Production Management*. https://doi.org/10.1108/IJOPM-08-2022-0478

Petersen, K., Vakkalanka, S., & Kuzniarz, L. (2015). Guidelines for conducting systematic mapping studies in software engineering: An update. *Information and Software Technology*, *64*, 1–18. https://doi.org/10.1016/j.infsof.2015.03.007

Pieroni, M. P. P., McAloone, T. C., & Pigosso, D. C. A. (2019). Business model innovation for circular economy and sustainability: A review of approaches. *Journal of Cleaner Production*, *215*, 198–216. https://doi.org/10.1016/j.jclepro.2019.01.036

Pisano, G. P. (1994). Knowledge, integration, and the locus of learning: An empirical analysis of process development. *Management Journal, 15*, 85–100.

Porter, M. E., & Kramer, M. R. (2006). Strategy & society: The link between competitive advantage and corporate social responsibility. *Harvard Business Review, 84*(12) *78–92*.Porter, M. E., & Van Der Linde, C. (1995). Toward a new conception of the environment-competitiveness relationship. *Journal of Economic Perspectives 9*(4), 97–118.

Potting, J., Hekkert, M., Worrell, E., & Hanemaaijer, A. (2017). Circular economy: Measuring innovation in the product chain—Policy report. In *PBL Netherlands Environmental Assessment Agency* (Issue 2544, p. 42).

Prahalad, C. K. (2004). The blinders of dominant logic. *Long Range Planning*, *37*(2), 171–179. https://doi.org/10.1016/j.lrp.2004.01.010

Raisch, S., & Birkinshaw, J. (2008). *Organizational ambidexterity: Antecedents*. https://doi.org/10.1177/0149206308316058

Rashid, A., Asif, F. M. A., Krajnik, P., & Nicolescu, C. M. (2013). Resource conservative manufacturing: An essential change in business and technology paradigm for sustainable manufacturing. *Journal of Cleaner Production*, *57*, 166–177. https://doi.org/10.1016/j.jclepro.2013.06.012

Reike, D., Vermeulen, W. J. V., & Witjes, S. (2018). The circular economy: New or Refurbished as CE 3.0? —Exploring Controversies in the Conceptualization of the Circular Economy through a Focus on History and Resource Value Retention Options. *Resources, Conservation and Recycling*, *135*, 246–264. https://doi.org/10.1016/j.resconrec.2017.08.027

Roome, N., & Louche, C. (2015). Journeying toward business models for sustainability: A conceptual model found inside the black box of organisational transformation. *Organization & Environment*, *29*. https://doi.org/10.1177/1086026615595084

Ruiter, H., De Feijter, F., & Wagensveld, K. (2022). Management control and business model innovation in the context of a circular economy in the Dutch construction industry. *Sustainability (Switzerland)*, *14*(1). https://doi.org/10.3390/su14010366

Saidani, M., Yannou, B., Leroy, Y., Cluzel, F., & Kendall, A. (2019). A taxonomy of circular economy indicators. *Journal of Cleaner Production*, *207*, 542–559. https://doi.org/10.1016/j.jclepro.2018.10.014

Santa-Maria, T., Vermeulen, W. J. V., & Baumgartner, R. J. (2021). Framing and assessing the emergent field of business model innovation for the circular economy: A combined literature review and multiple case study approach. *Sustainable Production and Consumption*, *26*, 872–891. https://doi.org/10.1016/j.spc.2020.12.037

Saunders, M., Lewis, P., & Thornhill, A. (2009). *Research methods for business students*. Pearson Education Limited.

Schaltegger, S., Hansen, E. G., & Lüdeke-Freund, F. (2016). Business models for sustainability: Origins, present research, and future avenues. *Organization and Environment*, *29*(1), 3–10. https://doi.org/10.1177/1086026615599806

Schaltegger, S., Lüdeke-Freund, F., & Hansen, E. G. (2012). Business cases for sustainability: The role of business model innovation for corporate sustainability. *International Journal of Innovation and Sustainable Development*, *6*(2), 95–119. https://doi.org/10.1504/IJISD.2012.046944

Seebode, D., Jeanrenaud, S., & Bessant, J. (2012). Managing innovation for sustainability. *R&D Management*, *42*(3), 195–206. https://doi.org/10.1111/j.1467-9310.2012.00678.x

Senge, P. M. (2006). *The fifth discipline: The art and practice of the learning organization* (2nd ed.). Doubleday/Currency.

Simons, R. (1995a). *Levers of control: How managers use innovative control systems to drive strategic renewal*. Harvard Business School Press.

Simons, R. (1995b, March 1). Control in an age of empowerment. *Harvard Business Review*. https://hbr.org/1995/03/control-in-an-age-of-empowerment

Smith, W. K., & Tushman, M. L. (2005). Managing strategic contradictions: A top management model for managing innovation streams. *Organization Science*, *16*(5), 522–536.

Song, X. M., & Montoya-Weiss, M. M. (1998). Critical development activities for really new versus incremental products. *Journal of Product Innovation Management*, *15*(2), 124–135. https://doi.org/10.1111/1540-5885.1520124

Sopjani, L., Arekrans, J., Laurenti, R., & Ritzén, S. (2020). Unlocking the linear lock-in: Mapping research on barriers to transition. *Sustainability*, *12*(3), 1034. https://doi.org/10.3390/su12031034

Spender, J.-C. (1996). Making knowledge the basis of a dynamic theory of the firm. *Strategic Management Journal*, *17*(S2), 45–62. https://doi.org/10.1002/smj.4250171106

Stahel, W. R. (2010). *The performance economy*. Palgrave Macmillan.

Stahel, W. R. (2019). *The circular economy: A user's guide*. Routledge.

Stonehouse, G., & Snowdon, B. (2007). Competitive advantage revisited: Michael Porter on strategy and competitiveness. *Journal of Management Inquiry*, *16*(3), 256–273. https://doi.org/10.1177/1056492607306333

Stringer, C. (2007). Empirical performance management research: Observations from AOS and MAR. *Qualitative Research in Accounting & Management*, *4*(2), 92–114. https://doi.org/10.1108/11766090710754187

Stubbs, W., & Cocklin, C. (2008). Conceptualizing a "sustainability business model." *Organization and Environment*, *21*(2), 103–127. https://doi.org/10.1177/1086026608318042

Sundin, H., Granlund, M., & Brown, D. A. (2010). Balancing multiple competing objectives with a balanced scorecard. *European Accounting Review*, *19*(2), 203–246. https://doi.org/10.1080/09638180903118736

Svensson, N., & Funck, E. K. (2019). Management control in circular economy. Exploring and theorizing the adaptation of management control to circular business models. *Journal of Cleaner Production*, *233*, 390–398. https://doi.org/10.1016/j.jclepro.2019.06.089

Tukker, A. (2004). Eight types of product-service system: Eight ways to sustainability? Experiences from suspronet. *Business Strategy and the Environment*, *260*, 246–260.

Tukker, A. (2015). Product services for a resource-efficient and circular economy—A review. *Journal of Cleaner Production*, *97*, 76–91. https://doi.org/10.1016/j.jclepro.2013.11.049

Tushman, M. L., & O'Reilly, C. A. (1996). Ambidextrous organizations: Managing evolutionary and revolutionary change. *California Management Review*, *38*(4), 8–29. https://doi.org/10.2307/41165852

Uhl-Bien, M., & Marion, R. (2009). Complexity leadership in bureaucratic forms of organizing: A meso model. *The Leadership Quarterly*, *20*(4), 631–650. https://doi.org/10.1016/j.leaqua.2009.04.007

Valant, J. (2016). *Planned obsolescence: Exploring the issue* (PE 581.999). European Parliamentary Research Service. https://www.europarl.europa.eu/RegData/etudes/BRIE/2016/581999/EPRS_BRI(2016)581999_EN.pdf

Van der Byl, C. A., & Slawinski, N. (2015). Embracing tensions in corporate sustainability: A review of research from win-wins and trade-offs to paradoxes and beyond. *Organization & Environment*, *28*(1), 54–79. https://doi.org/10.1177/1086026615575047

van der Meer-Kooistra, J., & Scapens, R. W. (2008). The governance of lateral relations between and within organisations. *Management Accounting Research*, *19*(4), 365–384. https://doi.org/10.1016/j.mar.2008.08.001

Veleva, V., & Bodkin, G. (2018). Corporate-entrepreneur collaborations to advance a circular economy. *Journal of Cleaner Production*, *188*, 20–37. https://doi.org/10.1016/j.jclepro.2018.03.196

von Bertalanffy, L. (1969). *General system theory: Foundations, development, applications*. George Braziller Inc.

Walsh, J. P. (1995). Managerial and organizational cognition: Notes from a trip down memory lane. *Organization Science*, *6*(3), 280–321.

Weick, K. E., Sutcliffe, K. M., & Obstfeld, D. (2005). Organizing and the process of sensemaking. *Organization Science*, *16*(4), 409–421. https://doi.org/10.1287/orsc.1050.0133

Wicks, A. C., & Freeman, R. E. (2008). Organization studies and the new pragmatism: Positivism, anti-positivism, and the search for ethics. *Organization Science*, *9*(2), 123–140. https://doi.org/10.1287/orsc.9.2.123

Yin, R. K. (2009). *Case study research: Design and methods*. SAGE Publications.

York, J. G. (2009). Pragmatic sustainability: Translating environmental ethics into competitive advantage. *Journal of Business Ethics*, *85*(1), 97–109. https://doi.org/10.1007/s10551-008-9950-6